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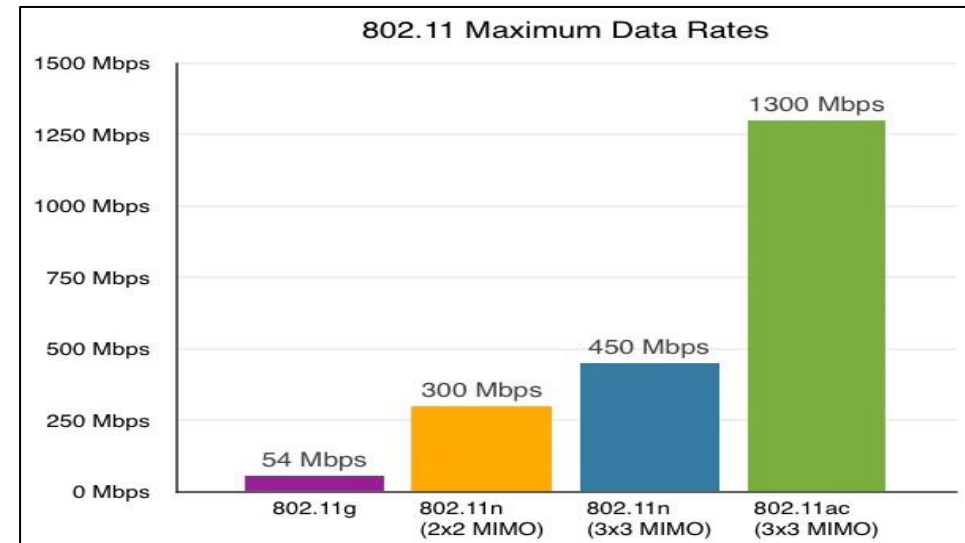
Orchestration of Network Slices in Fog, Edge, and Clouds

**Dr. B.V. Baiju,
SCOPE,
Assistant Professor
VIT, Vellore**

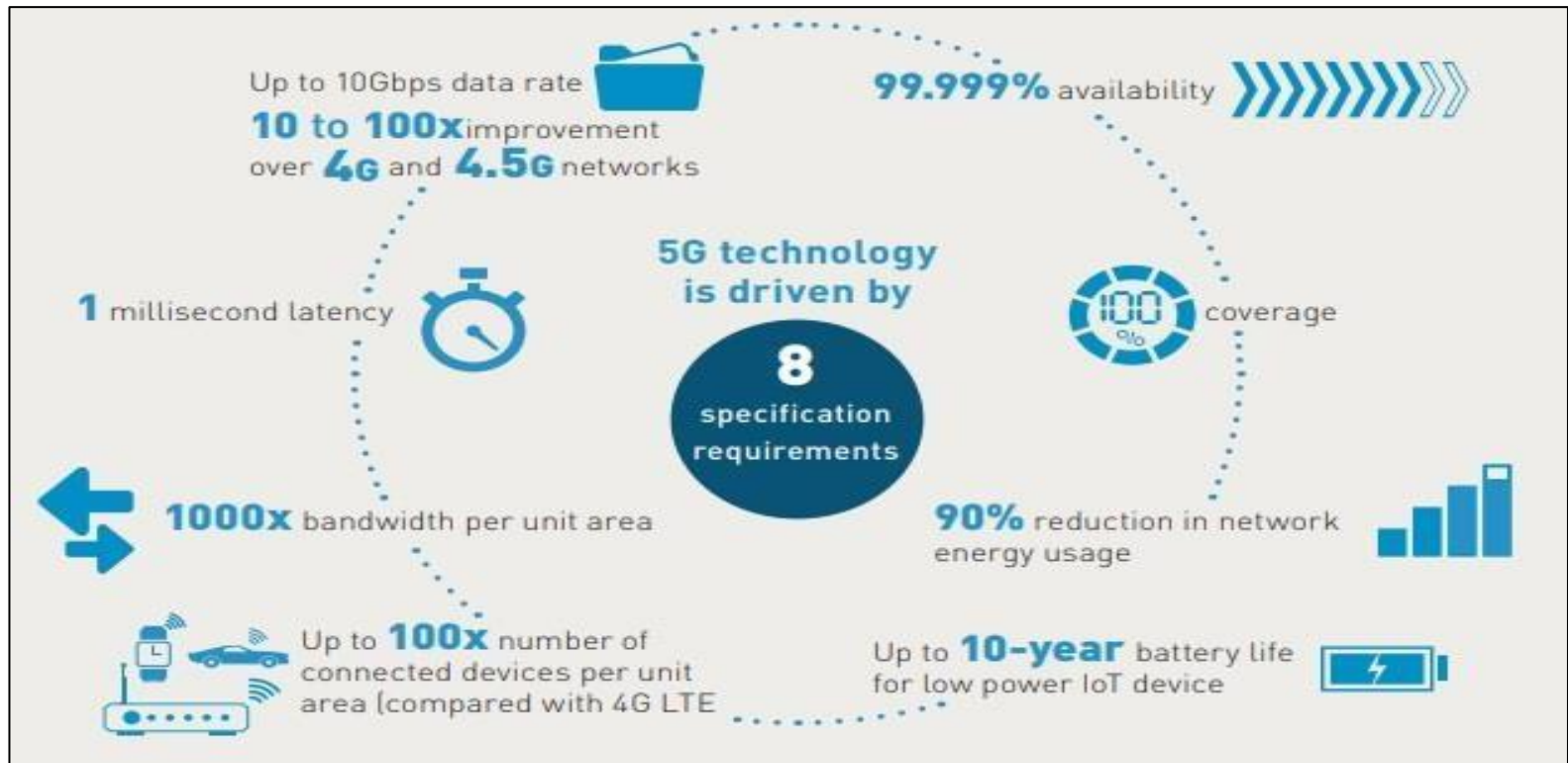
Background

a. 5G

- The wireless networking architecture of 5G follows **802.11ac**
- IEEE wireless networking criterion and operates on **millimeter wave bands**.
- It can encapsulate Extremely High Frequency (EHF) from 30 to 300 gigahertz (GHz) that ultimately offers higher data capacity and low latency communication.

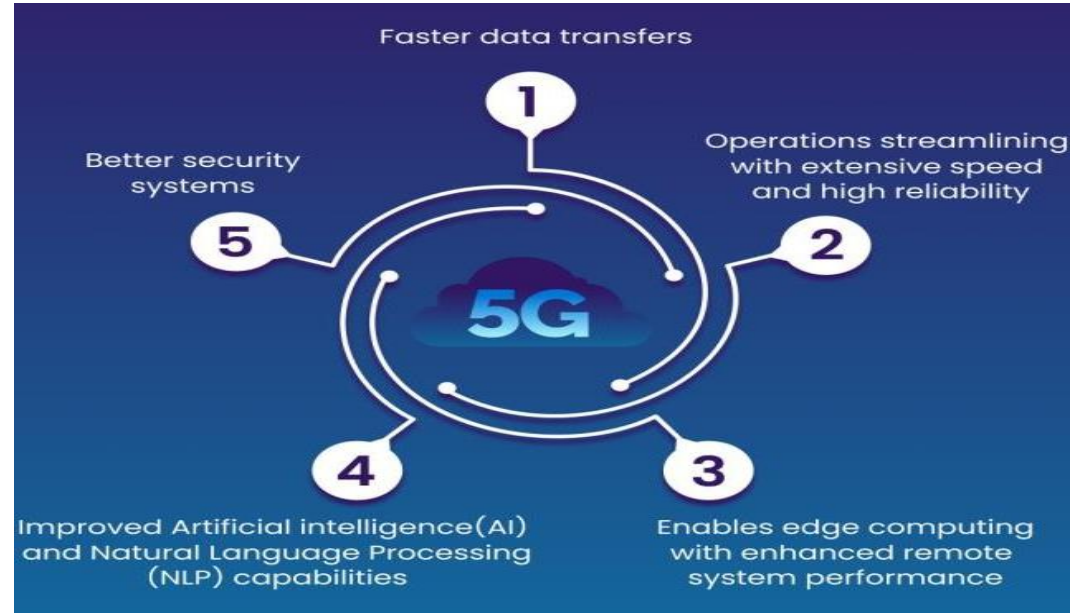


- The main intentions of 5G include enabling **Gbps data rate** in a real network with **least round-trip latency** and offering **long-term communication** among the large number of connected devices through high-fault tolerant networking architecture.
- 5G will be more **flexible**, **dynamic**, and **manageable** compared to the previous generations

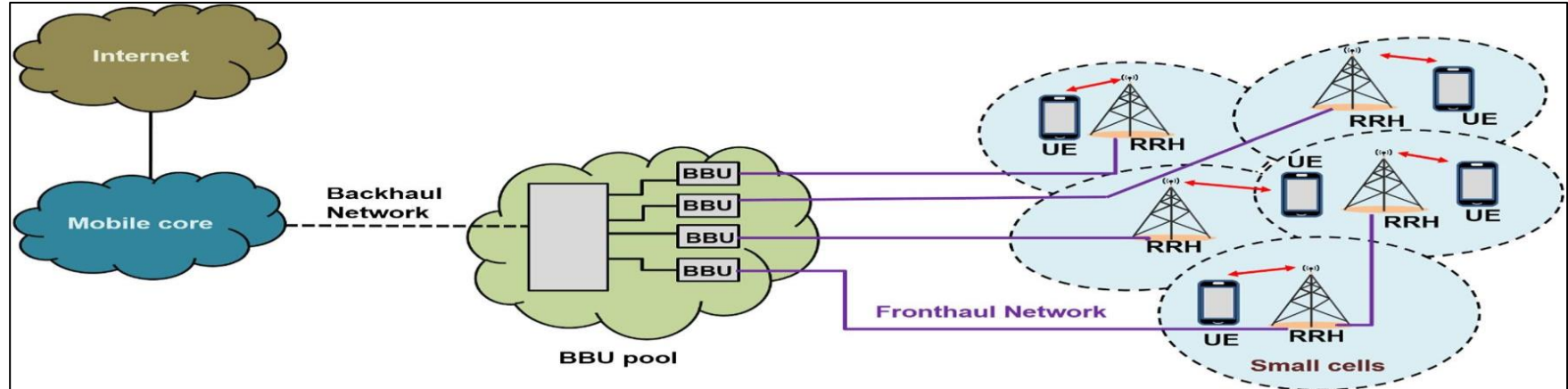


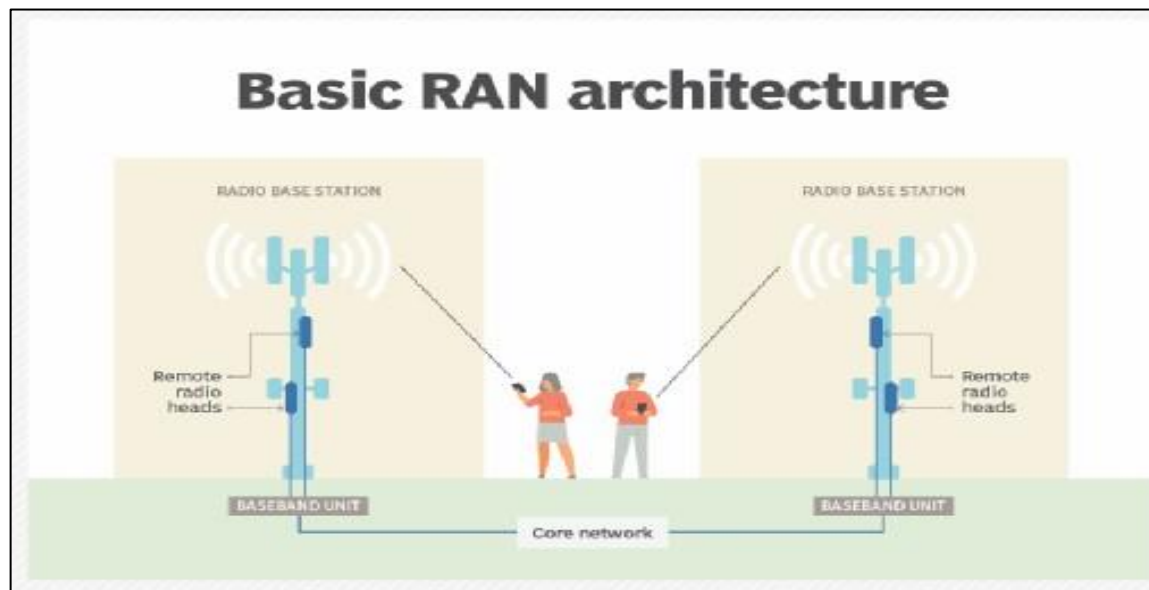
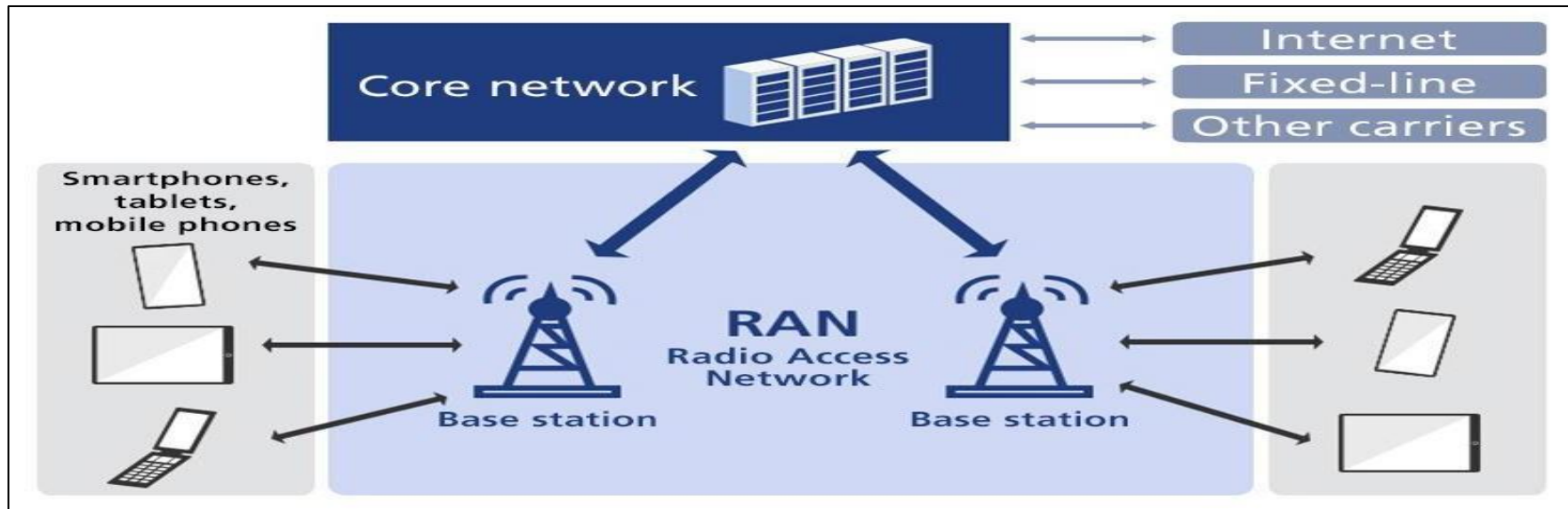
b. Cloud Computing

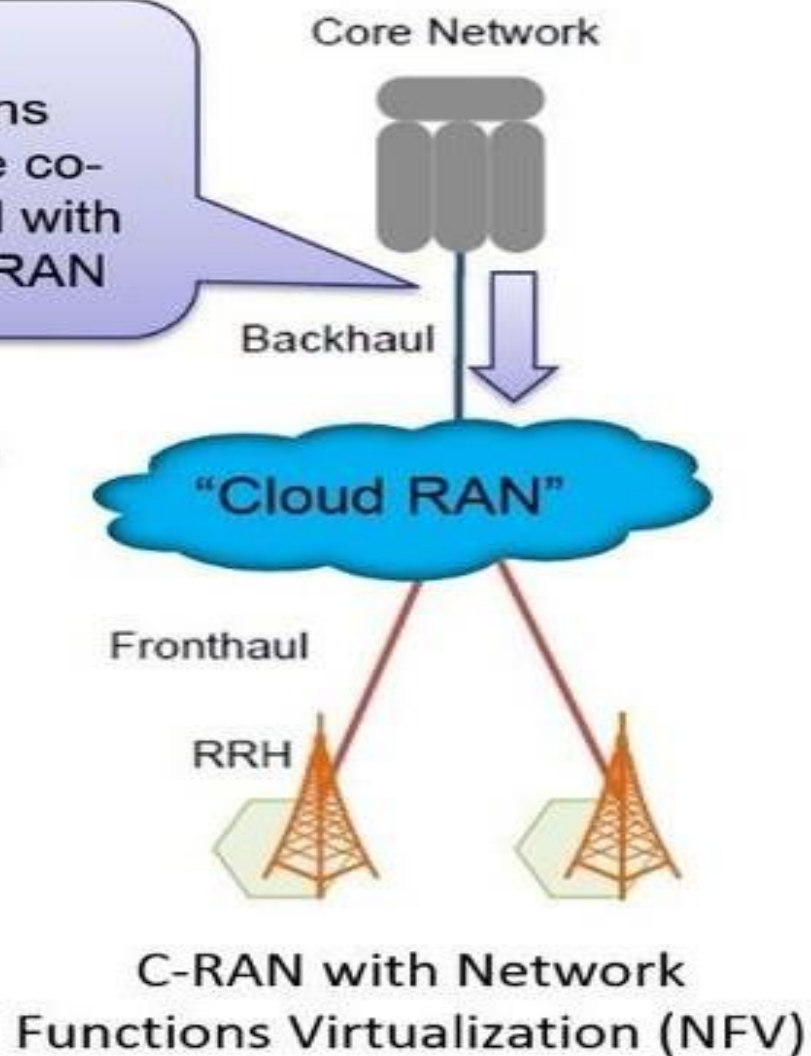
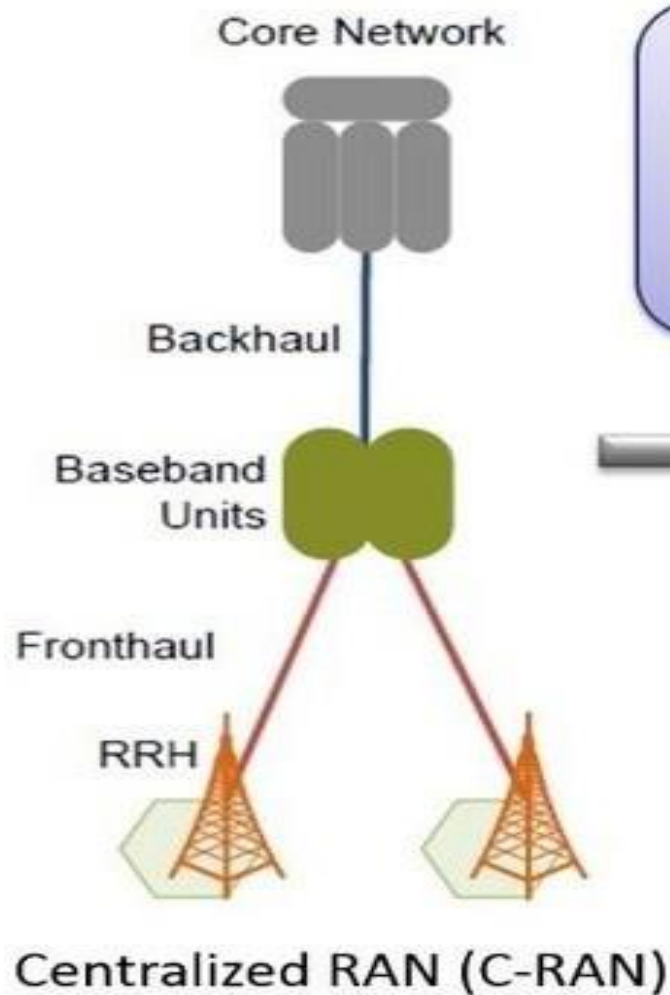
- Cloud computing is expected to be an inseparable part of 5G services for providing an excellent backend for applications running on the accessing devices.
- Cloud has evolved into a successful paradigm for computing delivering **on-demand services** over the Internet.
- The cloud adopted **technology** data centers **virtualization** for efficient management of resources and services.



- **SDC(Software Defined Clouds)** aims to utilize the advances in areas of **cloud computing**, **system virtualization**, **SDN**, and **NFV** to enhance resource management in data centers.
- Cloud is regarded as the foundation block for **Cloud Radio Access Network** (CRAN), an emerging cellular framework that aims to meet ever-growing end-user demand on 5G.
- In CRAN, the traditional base stations are split into radio and baseband parts.
- The **radio part** resides in the base station in the form of **the Remote Radio Head** (RRH) unit and the **baseband part** is placed to cloud for creating a centralized and virtualized **BaseBand Unit** (BBU) pool for different base stations.

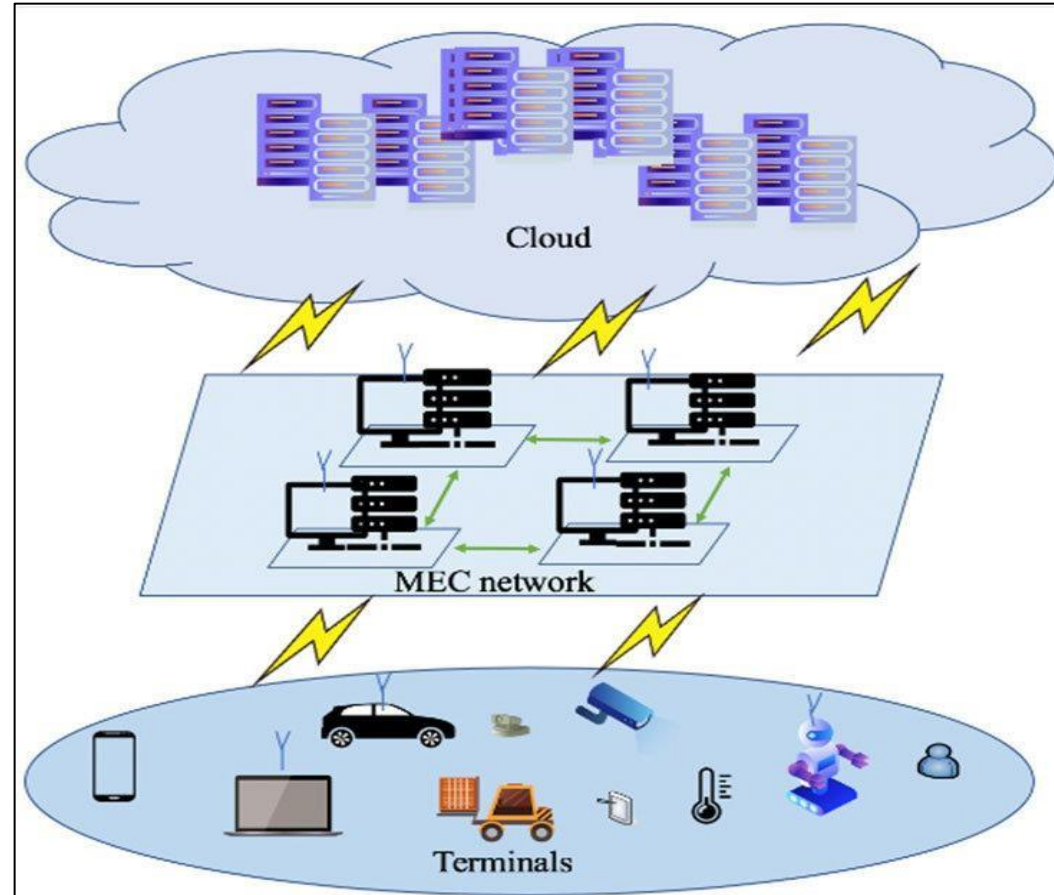






c. Mobile Edge Computing (MEC)

- MEC is considered as one of the key enablers of 5G.
- In MEC, base stations and access points are equipped with edge servers that take care of 5G-related issues at the edge network.
- MEC facilitates a enriched computationally distributed upon RAN architecture the LTE-based networking.

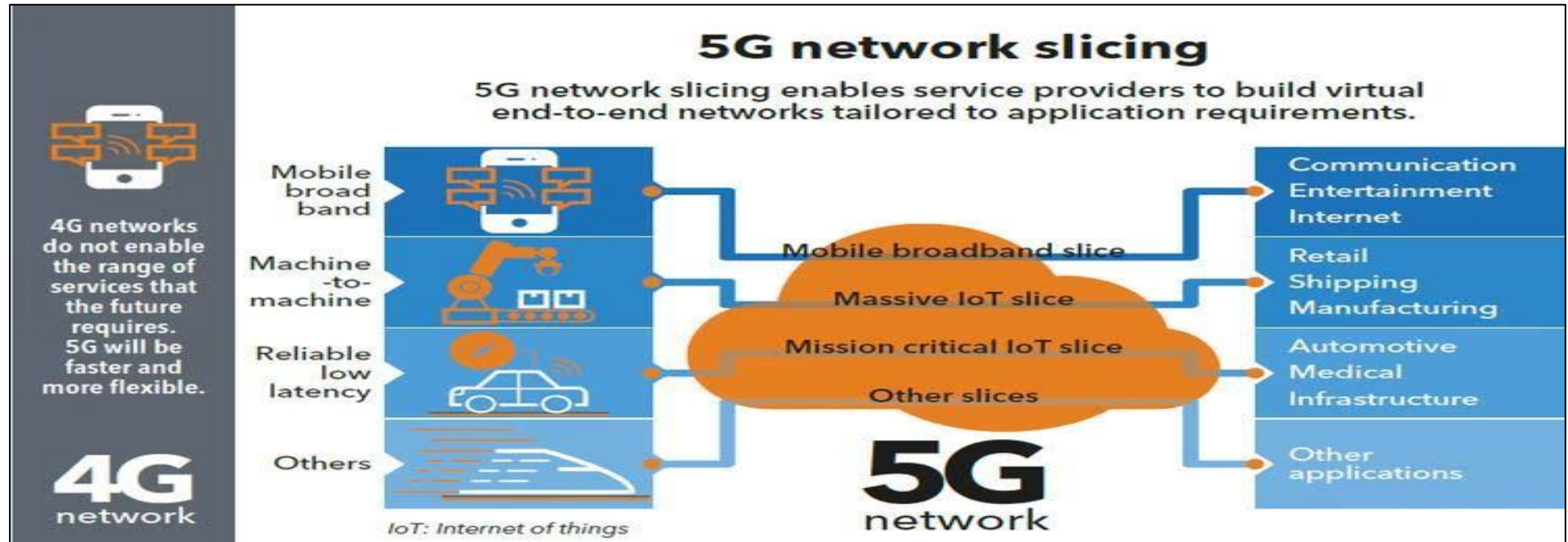


Edge and Fog Computing

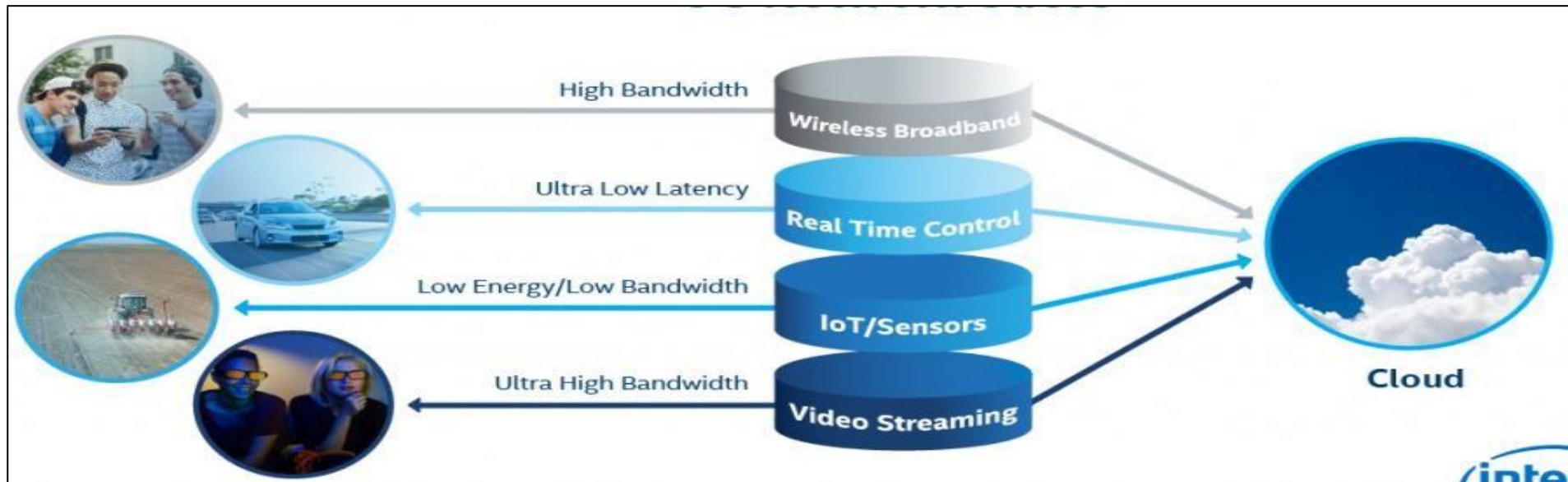
- Edge and fog computing are coined to complement the remote cloud to meet the service demands of a geographically distributed large number of IoT devices.

Network Slicing

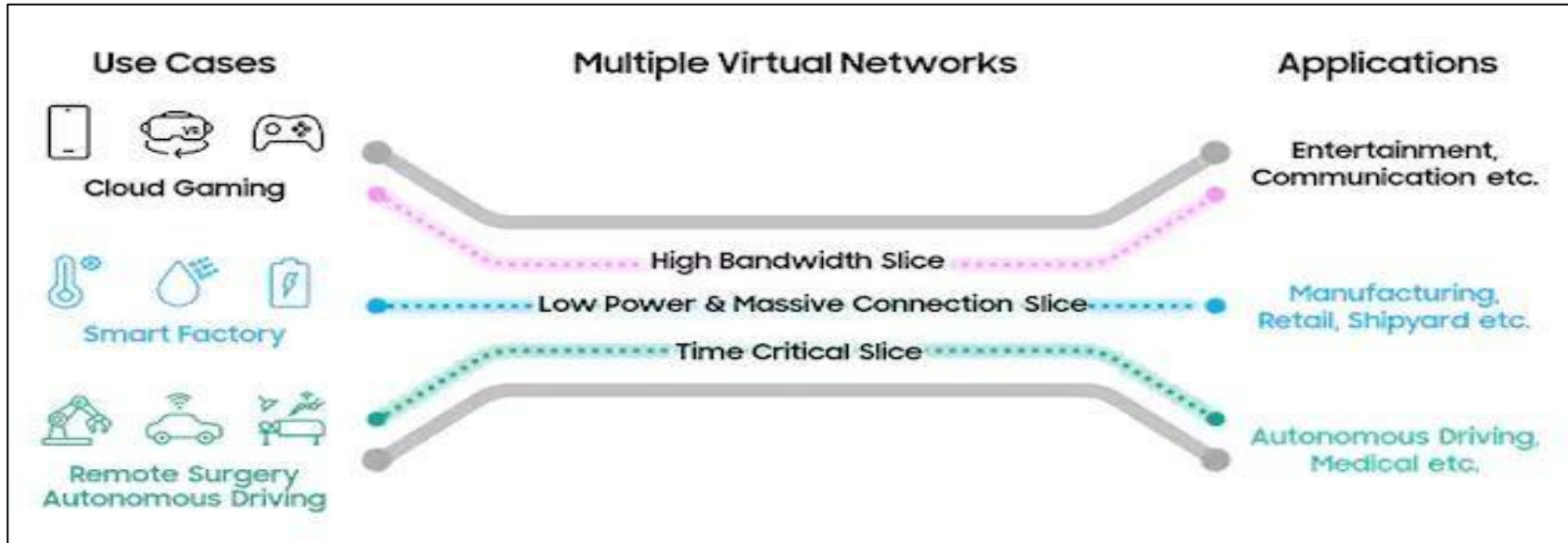
- Network slicing is a type of functionality that enables multiple independent networks to exist on the same physical network, using different “slices” of the same spectrum band.
- Network slicing in 5G refers to **sharing a physical network's resources** to multiple virtual networks



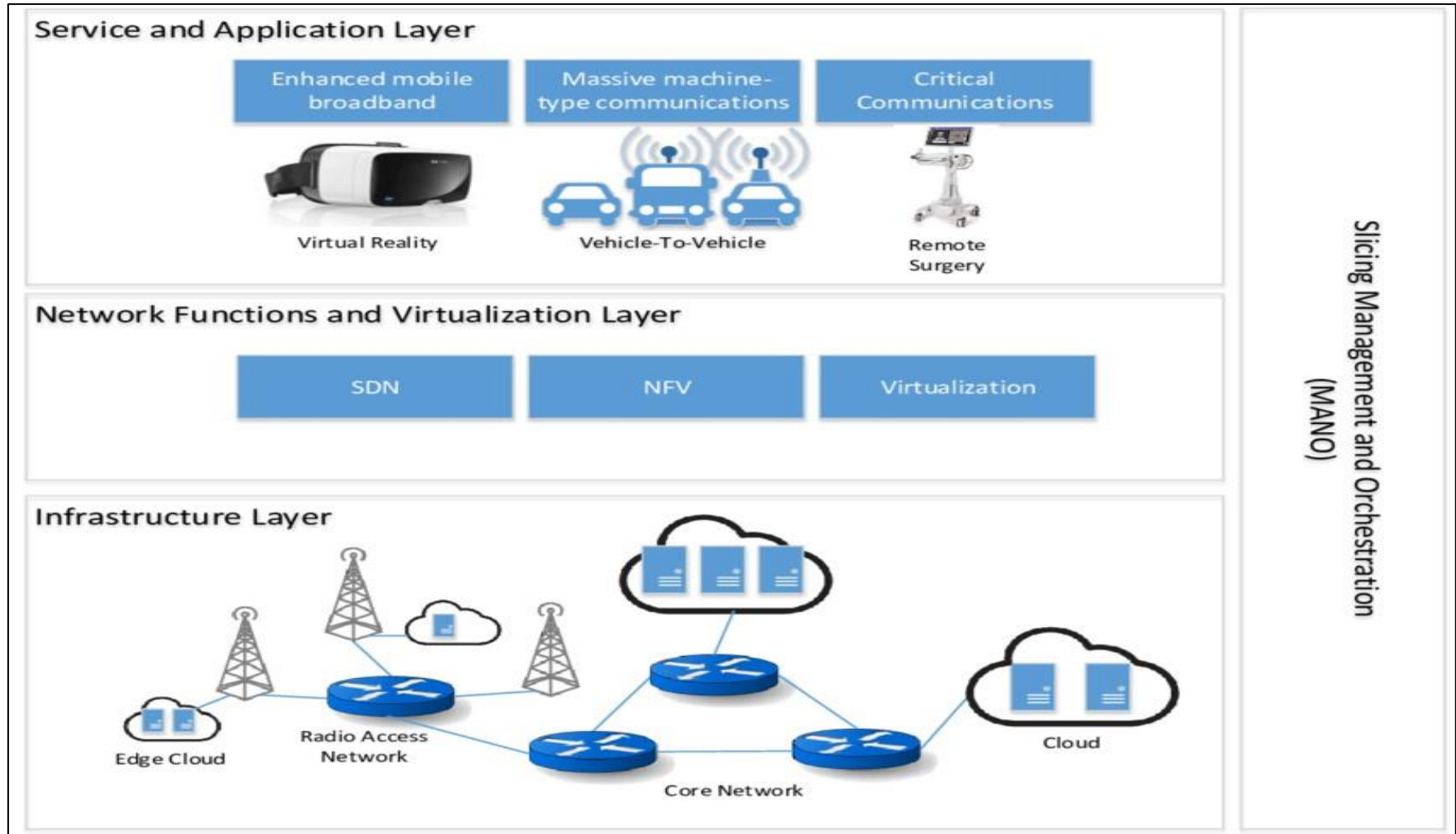
- Network slices are regarded as a **set of virtualized networks** on the top of a physical network.
- The network slices can be **allocated to specific applications/services, use cases or business models** to meet their requirements.
- Each network slice can be **operated independently** with its **own virtual resources, topology, data traffic flow, management policies, and protocols**.
- Network slicing usually requires implementation in an end-to-end manner to support coexistence of heterogeneous systems



- The network slicing paves the way for **customized connectivity** among a high number of interconnected end-to-end devices.
- Network slicing shares a common underlying infrastructure to **multiple virtualized networks**, it is considered as one of the most cost-effective ways to use network resources and reduce both capital and operational expenses.
- Network slicing assists **isolation and protection of data, control and management plane** that enforce security within the network.



- A generic framework for 5G network slicing consists of three main layers



1.1 Infrastructure Layer

- The infrastructure layer defines the actual physical network architecture.
- It can be expanded from **edge cloud to remote cloud** through **radio access network** and the **core network**.
- Different software defined techniques are encapsulated to facilitate resource abstraction within the core network and the radio access network.
- This layer **allocates resources** (compute, storage, bandwidth, etc.) to network slices in such way that upper layers can get access to handle them according to the context.

Consider a large office building with different departments: HR, IT, and Marketing. Each department has its own specific needs for internet usage.

Infrastructure Layer:

- This is like the building's physical network setup, including all the cables, routers, and servers. It extends from the local office (edge cloud) to the main data center (remote cloud).

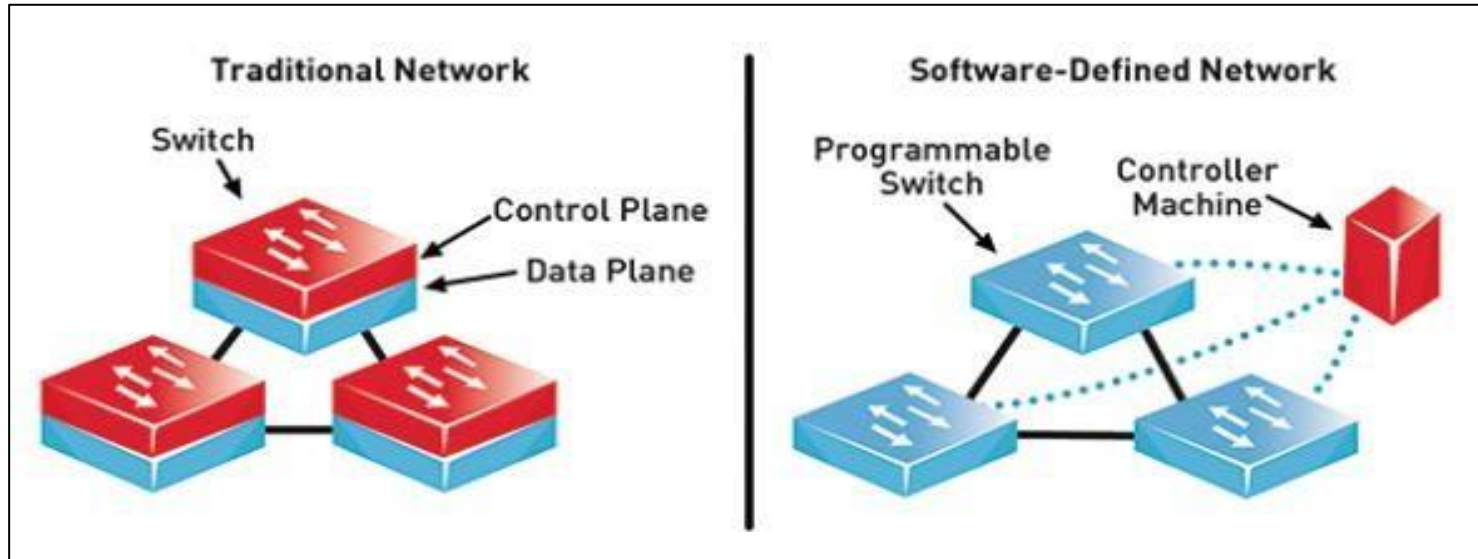
Resource Allocation:

The infrastructure layer allocates resources like internet bandwidth, storage, and computing power to each department based on their needs.

- **Network Slicing:**
 - **HR Department:** Needs stable and secure access for handling sensitive employee data. A dedicated slice ensures high security and reliability.
 - **IT Department:** Requires high-speed internet for software development and testing. A slice with high bandwidth and low latency is allocated.
 - **Marketing Department:** Uses a lot of cloud storage for media files and needs moderate internet speed. A slice with ample storage and moderate bandwidth is provided.
- **Software-Defined Techniques:**
 - These techniques manage and abstract the resources within the core network and radio access network, ensuring each department gets the right amount of resources without interfering with each other.

1.2 Network Function and Virtualization Layer

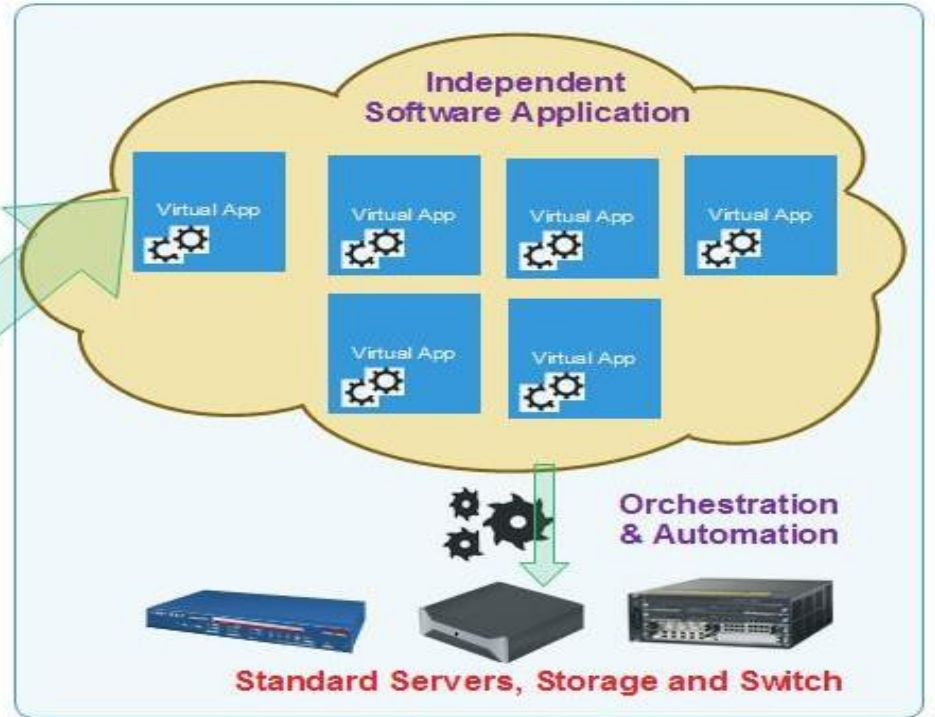
- The network function and virtualization layer executes all the required operations to manage the **virtual resources** and **network function's life cycle**.
- It also facilitates **optimal placement of network slices** to virtual resources and chaining of multiple slices so that they can meet specific requirements of a particular service or application.
- **SDN**, **NFV**, and different virtualization techniques are considered as the significant technical aspect of this layer.



Traditional Network Appliance Approach



Network Virtualisation Approach



- This layer explicitly manages the functionality of core and local radio access network.
- It can handle both **coarse-grained** and **fine-grained** network functions efficiently.

Smart City Infrastructure:

Scenario: **A city implements a smart city network to manage various services like traffic control, public safety, and utility management.**

Coarse-Grained Slicing: The city creates broad network slices for each major service:

- **Traffic Control Slice:** Allocates resources to manage traffic lights, sensors, and cameras across the city.
- **Public Safety Slice:** Dedicates resources for emergency services, surveillance cameras, and communication systems for police and fire departments.
- **Utility Management Slice:** Manages resources for water, electricity, and waste management systems.

Autonomous Vehicles:

Scenario: **Within the traffic control slice, there are autonomous vehicles that require real-time data for safe operation.**

Fine-Grained Slicing: Traffic control slice further subdivides into more specific slices:

- **Vehicle-to-Infrastructure (V2I) Communication Slice:** Ensures low-latency communication between vehicles and traffic signals.
- **Vehicle-to-Vehicle (V2V) Communication Slice:** Allocates resources for direct communication between autonomous vehicles to avoid collisions.
- **High-Definition Mapping Slice:** Provides high-bandwidth resources for real-time map updates and navigation.

1.3 Service and Application Layer

- The service and application layer can be composed by **connected vehicles**, **virtual reality appliances**, **mobile devices**, etc. having a specific use case or business model and represent certain utility expectations from the networking infrastructure and the network functions.



1. 4 Slicing Management and Orchestration (MANO)

- The functionality of the above layers are explicitly monitored and managed by the slicing management and orchestration layer.
- There are three main tasks in this layer:

Create virtual network instances upon the physical network by using the functionality of the infrastructure layer



Map network functions to virtualized network instances to build a service chain with the association of network function and virtualization layer.

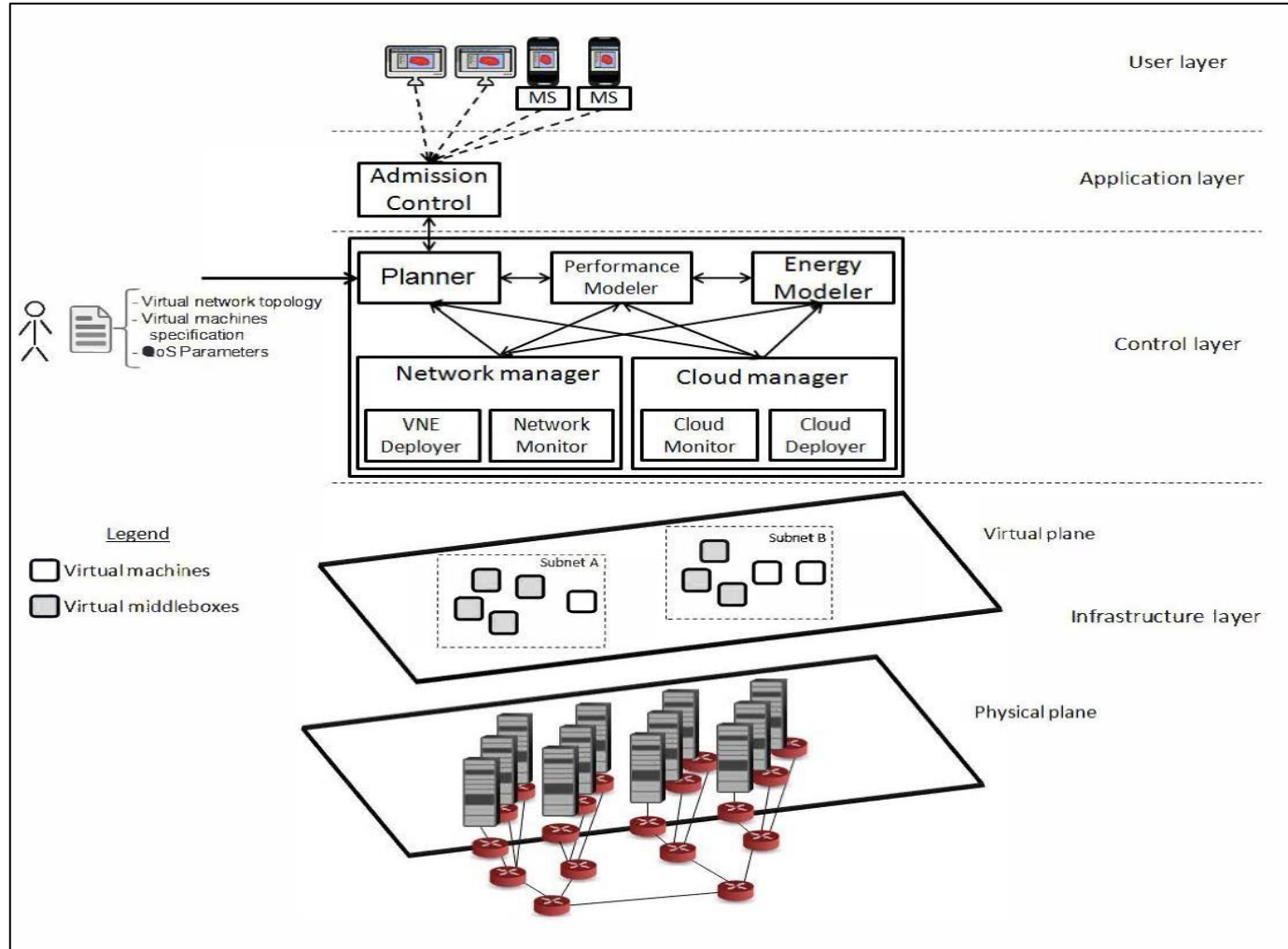


Maintain communication between service/application and the network slicing framework to manage the life cycle of virtual network instances and dynamically adapt or scale the virtualized resources according to the changing context.

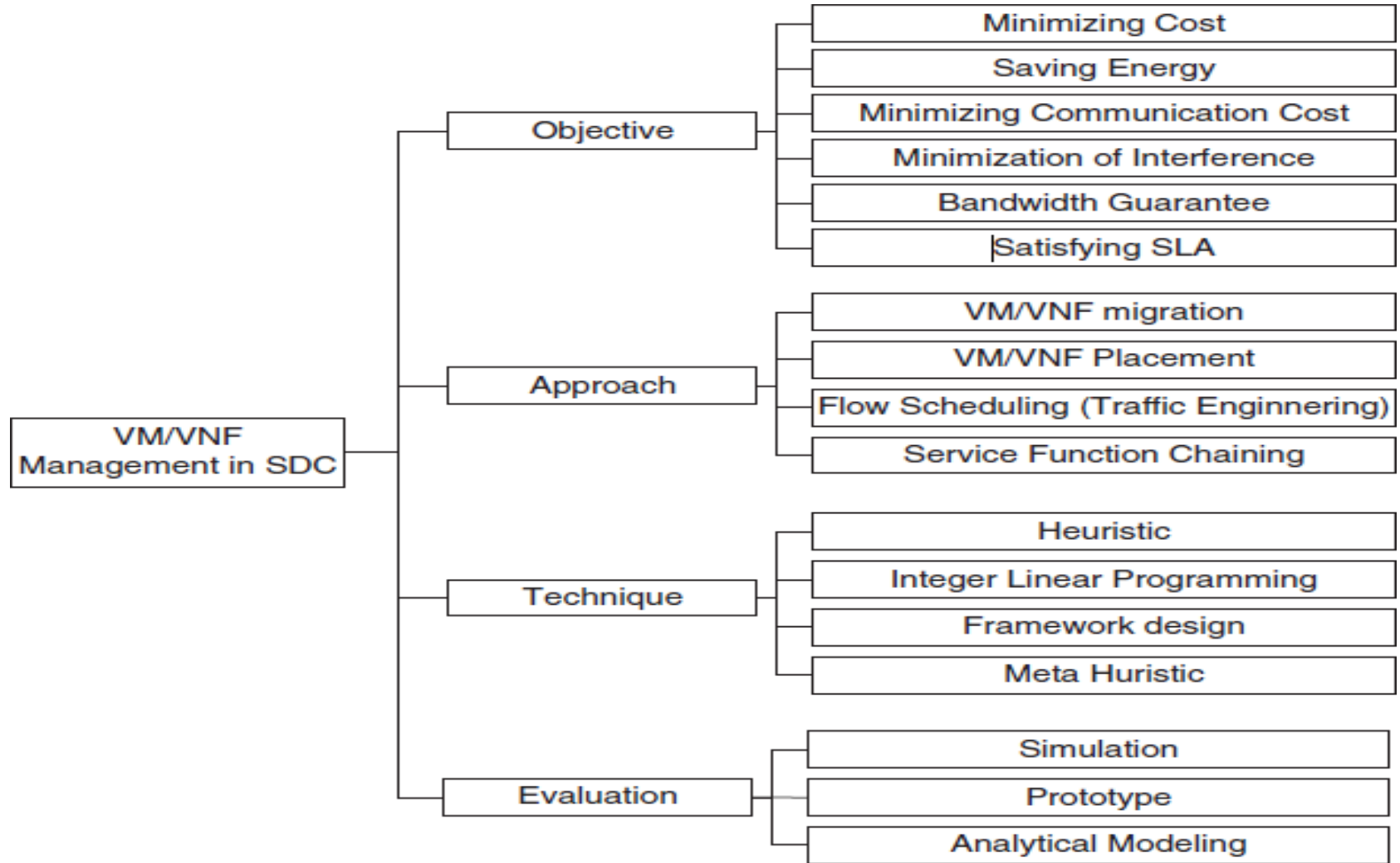
- A high-level perspective of 5G network, **Cloud-Native network architecture** for 5G has four characteristics:
 1. It provides **cloud data center-based architecture** and logically independent network slicing on the network infrastructure to support different application scenarios.
 2. It uses Cloud-RAN1 to build radio access networks (RAN) to provide a substantial number of connections and implement 5G required on-demand deployments of RAN functions.
 3. It provides simpler core network architecture and provides on- demand configuration of network functions via user and control plane separation, unified database management, and component- based functions.
 4. In an automatic manner, it implements network slicing service to reduce operating expenses.

Network Slicing in Software-Defined Clouds

- Software Defined Clouds (SDC) as an approach to **automate the Optimal cloud configuration process** by extending the concept of virtualization to all resources in a data center.



- The below figure illustrates the proposed taxonomy of network-aware VM/VNF management in SDCS.

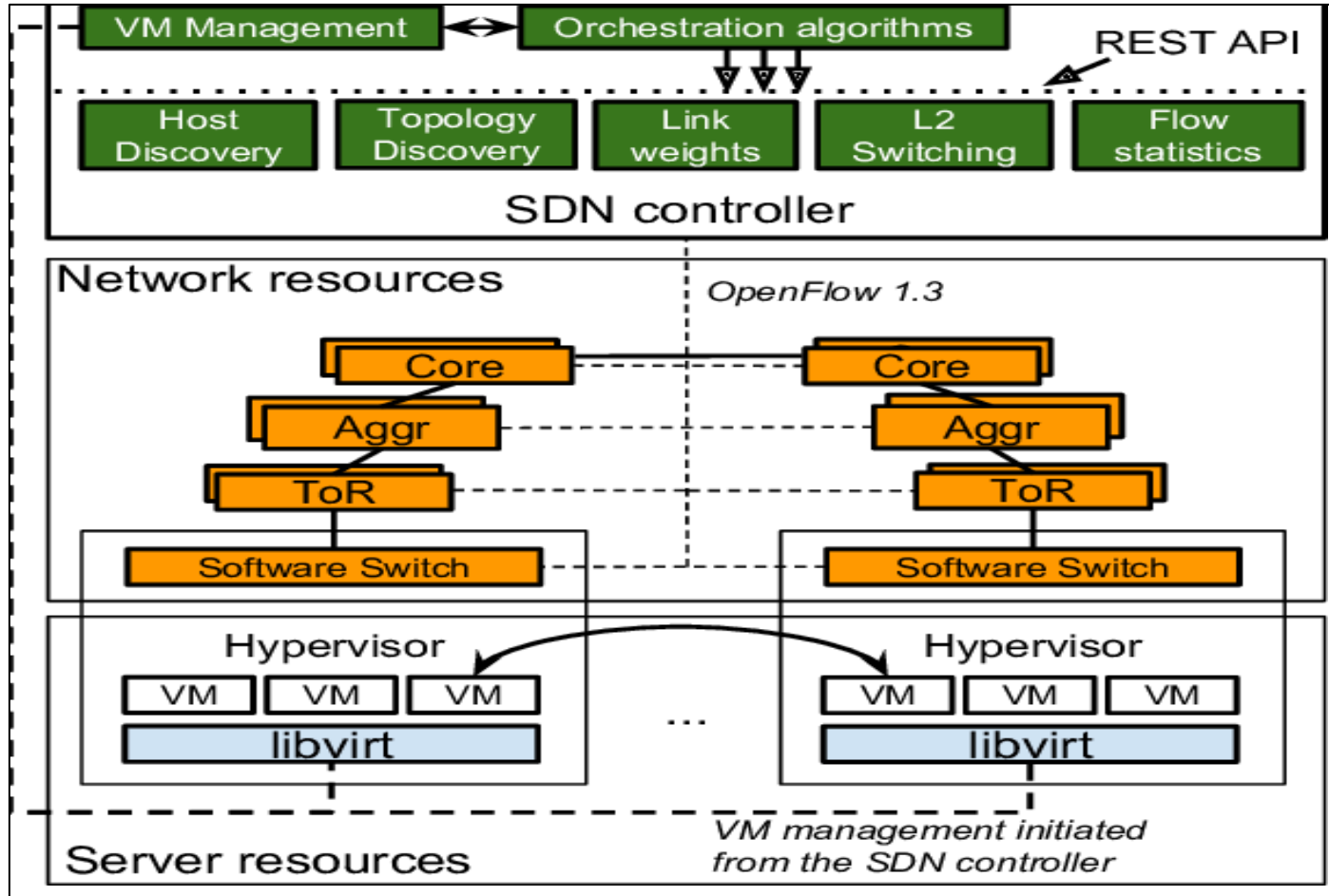


a.Network-Aware Virtual Machines Management

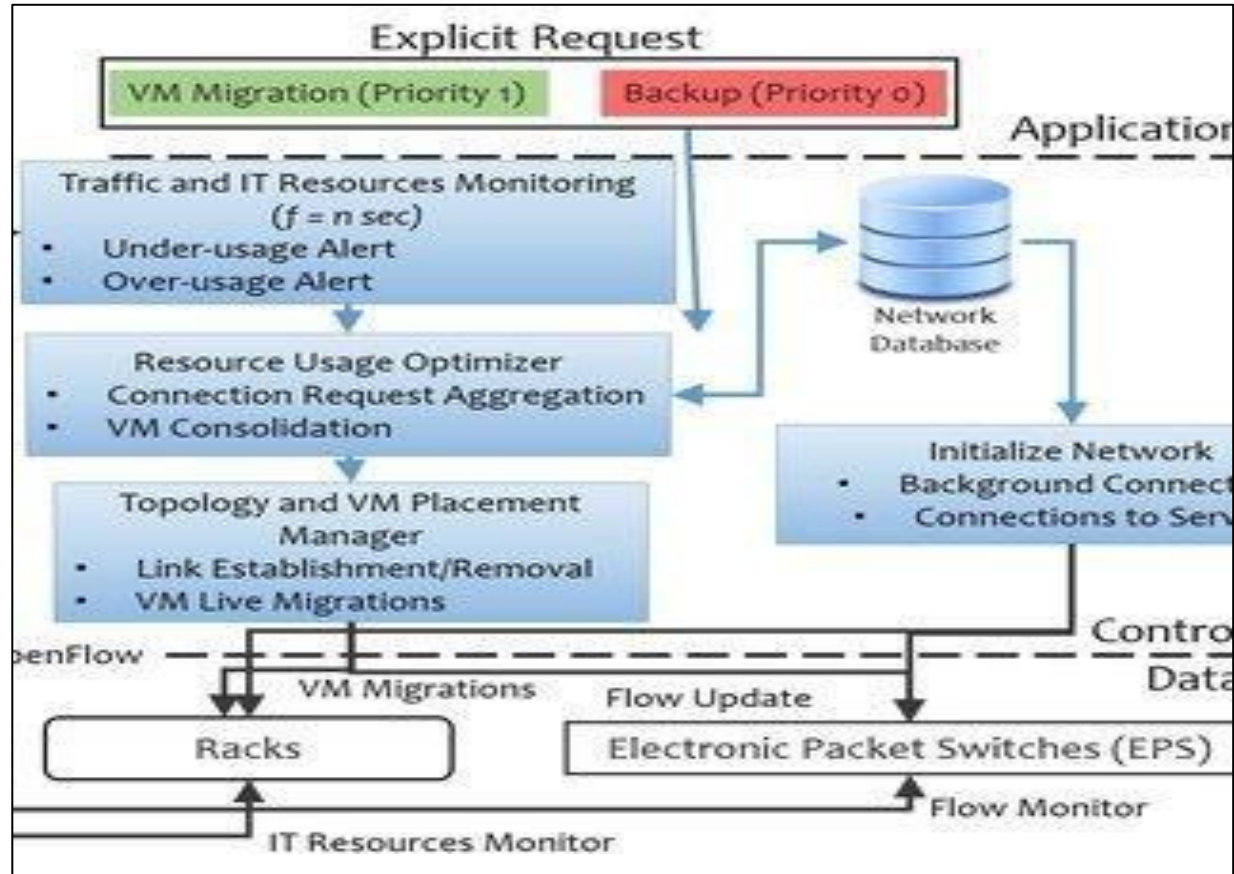
- Table summarizes the research projects on network-aware VM management.

Project	Objectives	Approach/Technique	Evaluation
Cziva et al. [29]	Minimization of the network communication cost	VM migration – Framework Design	Prototype
Wang et al. [30]	Reducing the number of hops between communicating VMs and network power consumption	VM placement – Heuristic	Simulation
Remedy [31]	Removing congestion in the network	VM migration – Framework Design	Simulation
Jiang et al. [32]	Minimization of the network communication cost	VM Placement and Migration – Heuristic (Markov approximation)	Simulation
VMPlanner [33]	Reducing network power consumption	VM placement and traffic flow routing - Heuristic	Simulation
PLAN [35]	Minimization of the network communication cost while meeting network policy requirements	VM Placement - Heuristic	Prototype/ Simulation

- **Cziva et al.** (SDN-Based Virtual Machine Management for Cloud Data Centers) present an **orchestration framework** to exploit **time-based network information** to live migrate VMs and minimize the network cost.



- Wang et al. (EQVMP: Energy-efficient and QoS-aware virtual machine placement for software defined datacenter networks)** proposed a **VM placement mechanism** to *reduce the number of hops* between communicating VMs, save energy, and balance the network load.
- Vijay Mann et al. (Remedy: Network-aware steady state VM management for data centers)** relies on **SDN** to monitor the state of the network and estimate the cost of VM migration.
 - Their technique **detects congested links** and migrates VMs to remove congestion on those links.



- **Jiang et al. (Joint VM Placement and Routing for Data Center Traffic Engineering)** worked on **joint VM placement** and **network routing problem** of data centers to minimize network cost in real-time.
 - They proposed an online algorithm to **optimize the VM placement** and **data traffic routing** with dynamically adapting traffic loads.

VMPR-online

Upon job k arrival:

$\mathcal{G}_k \leftarrow \text{GreedyAdd}(G, f_{k-1}, k)$

$\mathcal{F}_k \leftarrow \{f_{k-1}\} \times \mathcal{G}_k$

Choose $f \in \mathcal{F}_k$ with probability proportional to $\exp(-\beta x_f)$

Accommodate job k and set up routing according to f

$f_k \leftarrow f$

Upon job k departure:

Find the most-congested edge $e_l \in E(G)$

Let J be the set of jobs that use edge e_l

for each job $j \in J$

Let $\mathcal{F}_j$ be the set of possible configurations by migrating j

Calculate the transition probability of $f_j \in \mathcal{F}_j$ acc. to (19)

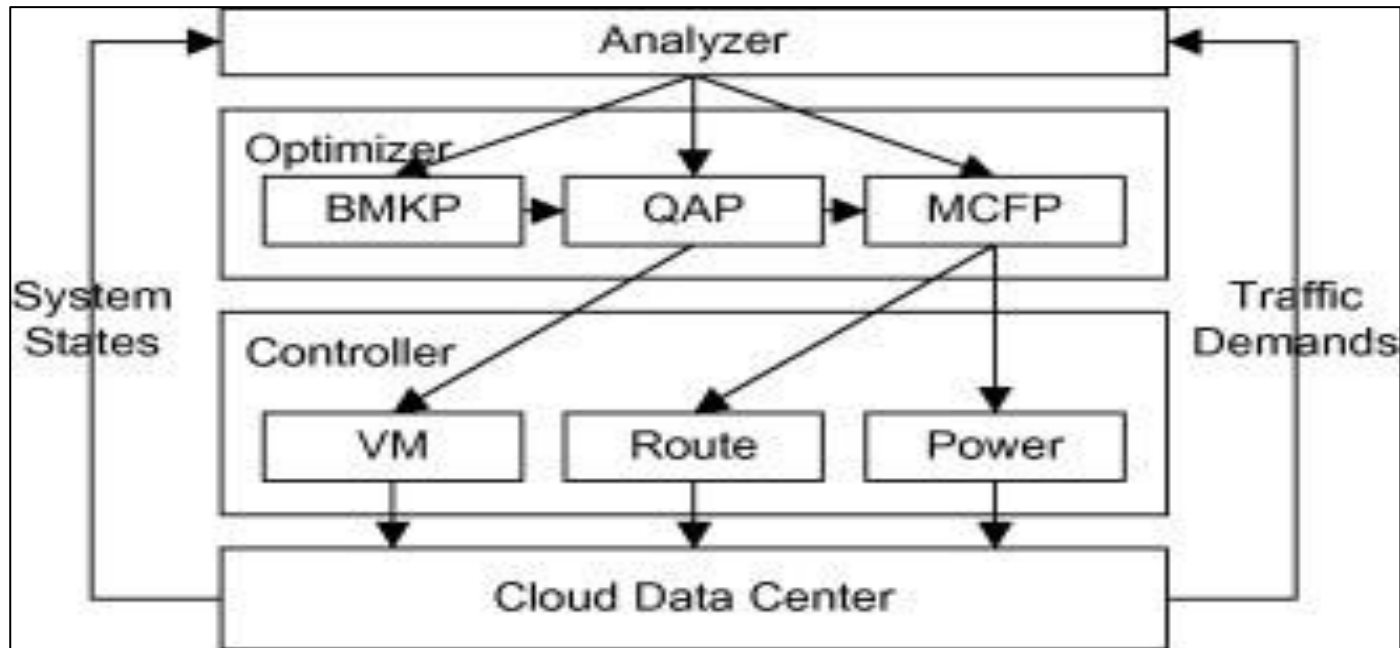
end for

$\mathcal{F}_{k-1} \leftarrow \bigcup_{j \in J} \mathcal{F}_j$

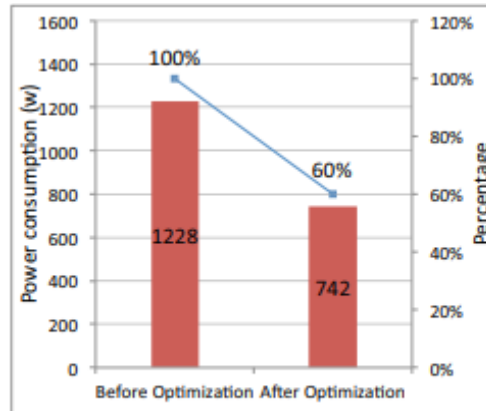
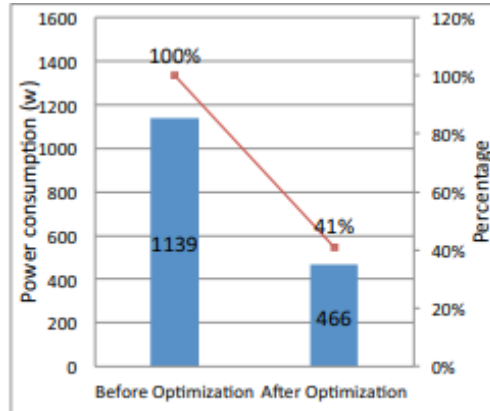
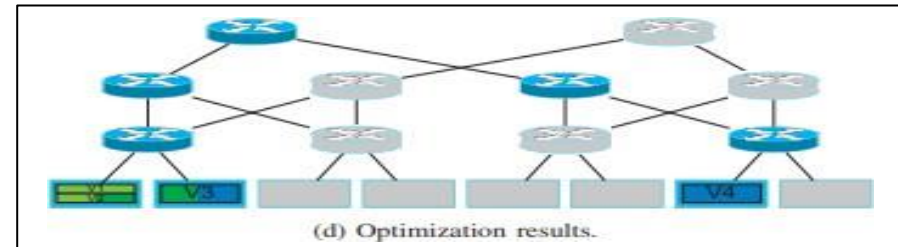
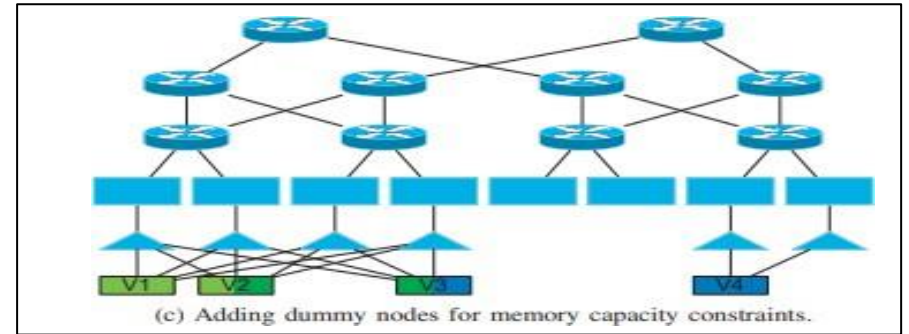
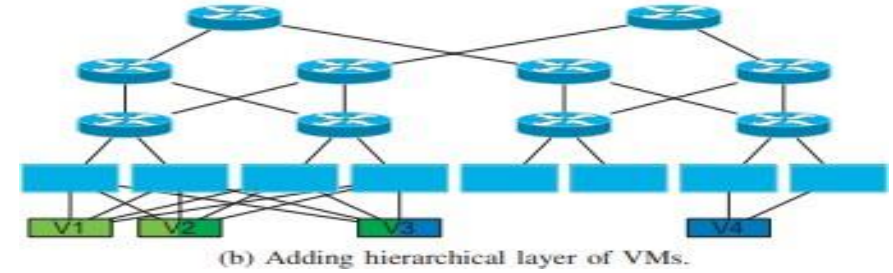
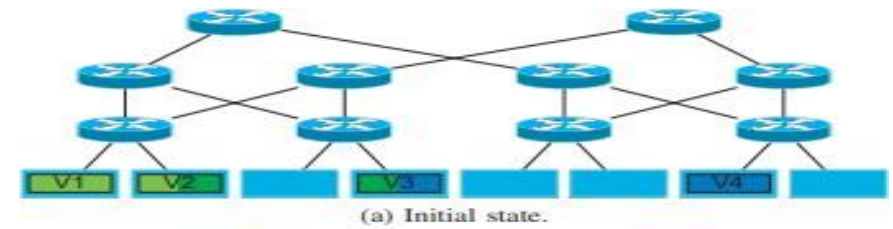
Migrate to configuration $f \in \mathcal{F}_{k-1}$ probabilistically

$f_{k-1} \leftarrow f$

- **Fang et al.** (VMPlanner: Optimizing virtual machine placement and traffic flow routing to reduce network power costs in Cloud data centers) also optimizes VM placement and network routing.
- The solution includes VM grouping that consolidates
 - **VMs with high inter-group traffic**
 - **VM group placement within a rack**
 - **Traffic consolidation to minimize the rack traffic.**

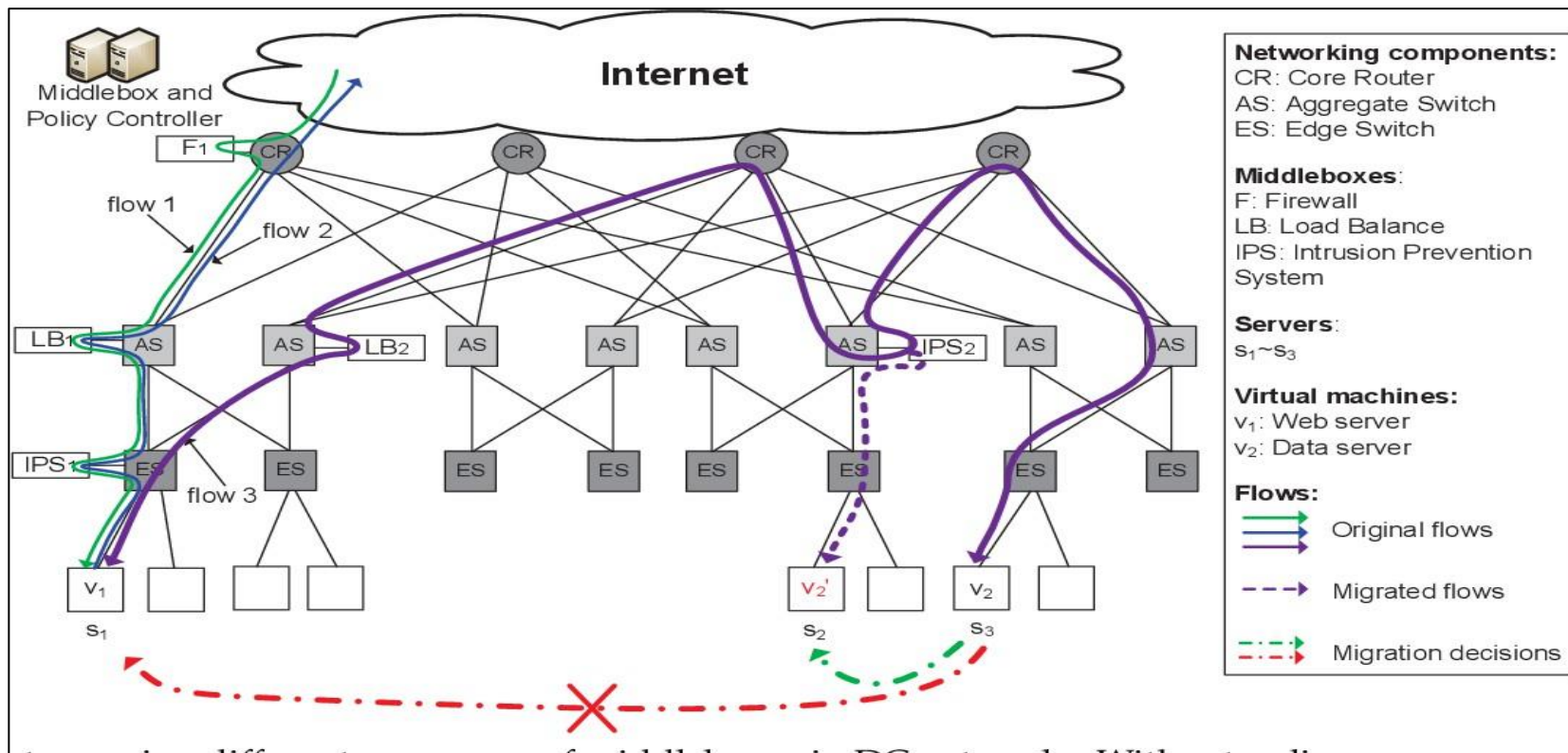


- **Jin et al. (Joint host-network optimization for energy-efficient data center Networking)** studied **joint host-network optimization problem**. The problem is formulated as an integer linear problem that combines VM placement and routing problem.



(a) Memory and traffic load = 15%. (b) Memory and traffic load = 30%.

- **Cui et al. (Joint policy- and network-aware VM management for cloud data centers)** explore the **joint policy-aware** and **network-aware VM** migration problem and present a VM management to reduce network-wide communication cost in data center networks while considering the policies regarding the network functions and middle boxes.



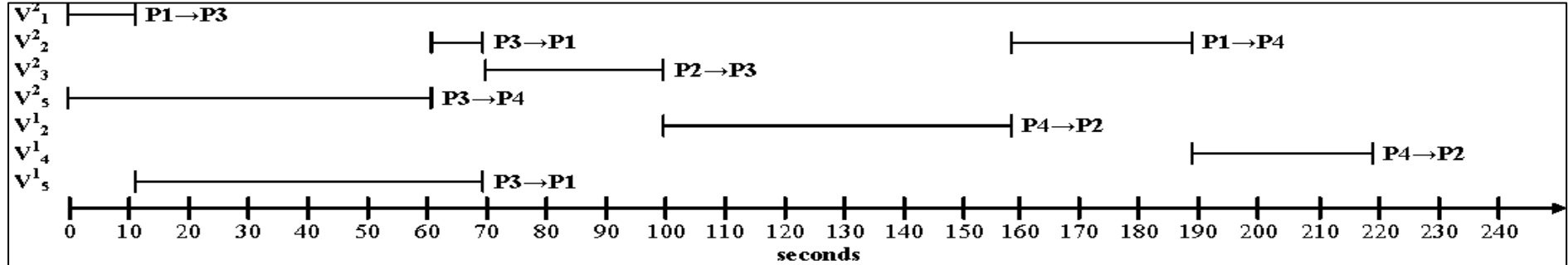
b. Network-Aware Virtual Machine Migration Planning

- **Bari et al. (CQNCR: Optimal VM migration planning in cloud data centers)** proposed a method for finding an **efficient migration plan**.
- They try to find a sequence of migrations to move a group of VMs to their final destinations while migration time is minimized.

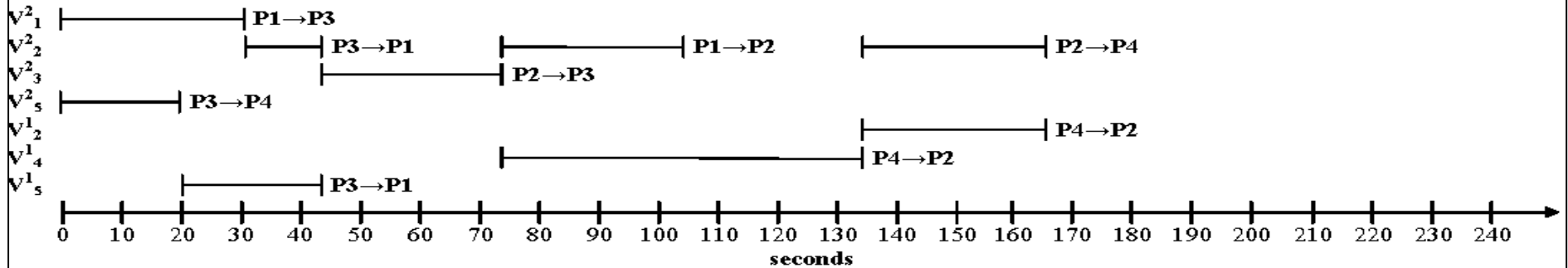
Imagine you are organizing a school event and need to move several groups of students (VMs) from classrooms (servers) to specific activity zones (final destinations).

- ***Plan the Moves:*** They figure out the best order to move the students so the process is smooth and doesn't cause delays.
- ***Minimize Time:*** They ensure students take the shortest routes and avoid traffic in the hallways (network congestion).
- ***Group Migration:*** Instead of moving everyone at once and causing chaos, they move smaller groups in an efficient sequence to save time.

- The method involves *tracking the remaining bandwidth* on the network links between the source and destination after each step in the sequence of operations.



(a) Sequence 1



(b) Sequence 2

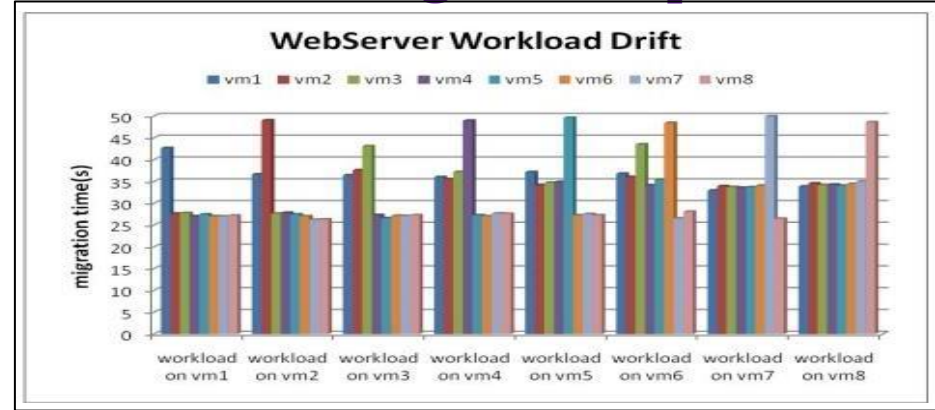
Fig. 4. VM migration sequences

- **Ghorbani et al. (Walk the line: Consistent network updates with bandwidth guarantees)** propose an algorithm to generate an ordered list of VMs to migrate and a set of forwarding flow changes.
- They focus on ensuring that the *links have enough bandwidth* so their capacity isn't exceeded during the migration.

Imagine you have a road with limited space, and several trucks need to move from one city to another without causing traffic jams.

- *Order the Trucks (VMs):* They decide the sequence in which trucks should move to avoid congestion.
- *Check Road Capacity (Bandwidth):* They ensure there's enough space on the road so trucks can move without overcrowding.
- *Update Traffic Signals (Flow Changes):* They adjust signals to guide trucks efficiently while keeping the road clear.

- **Li et al.** (Informed live migration strategies of virtual machines for cluster load balancing) tackled the **VM migration planning problem** where they address the **workload-aware migration problem** and propose methods for selection of
 - **candidate virtual machines**
 - **destination hosts**
 - **sequence for migration.**



Imagine you have several classrooms (servers) with students (VMs), and some classrooms are overcrowded while others have space.

Select Candidate Students (VMs): Identify which students need to move to reduce overcrowding. For example, pick students from the fullest classrooms.

Find Destination Classrooms (Hosts): Choose classrooms with enough free space where these students can be relocated without overcrowding.

Plan the Sequence: Decide the order in which students should move to ensure a smooth process without creating new overcrowding along the way.

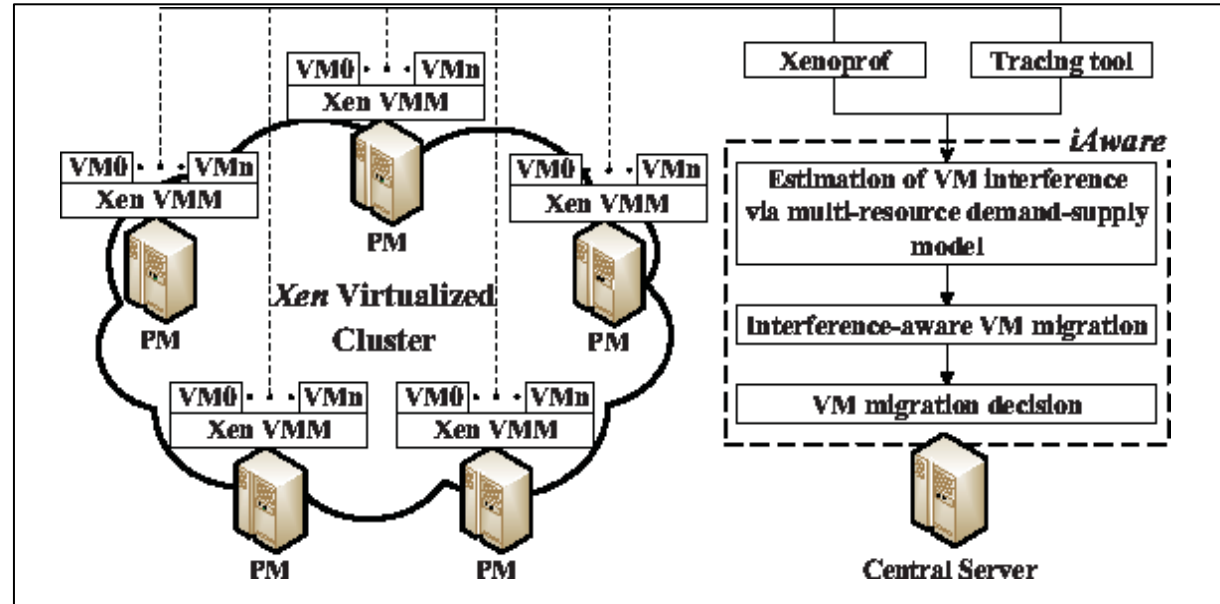
- **Xu et al.** (**iAware: Making Live Migration of Virtual Machines Interference-Aware in the Cloud**) propose an **interference-aware VM live migration plan** called ***iAware*** that minimizes both migration and co-location interference among VMs.

Migration Interference:

- During live migration, the movement of VMs consumes network and CPU resources. This can lead to degraded performance for both the migrating VM and other VMs sharing the same physical host or network.

Co-location Interference:

- When multiple VMs are hosted on the same physical server, they may compete for shared resources like CPU, memory, and storage, causing performance degradation due to resource conflict.

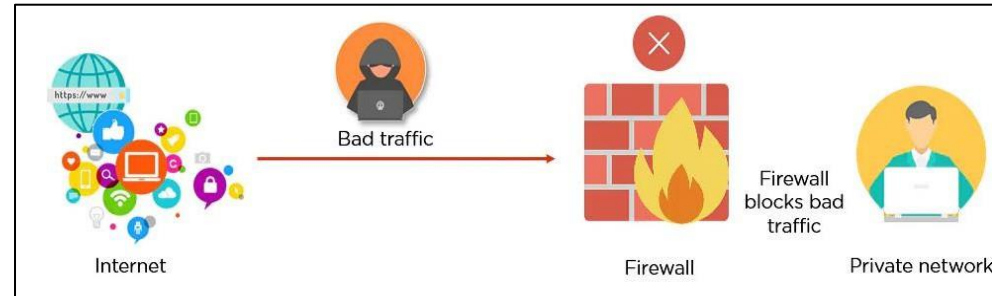
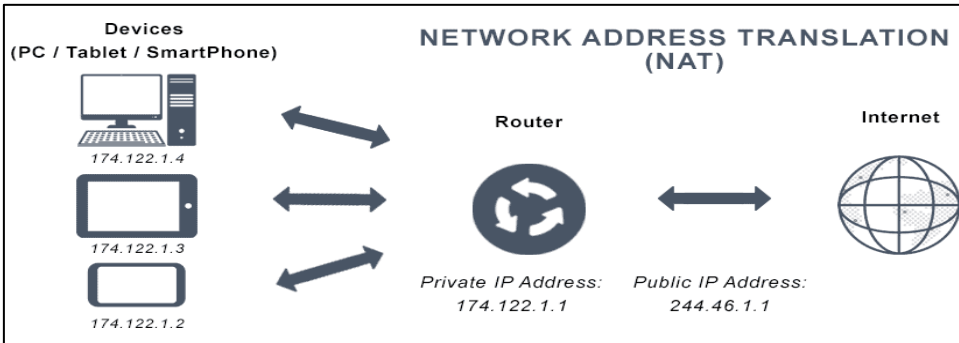
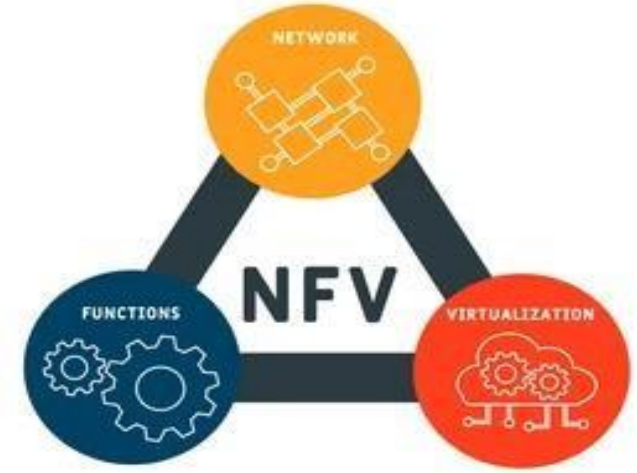
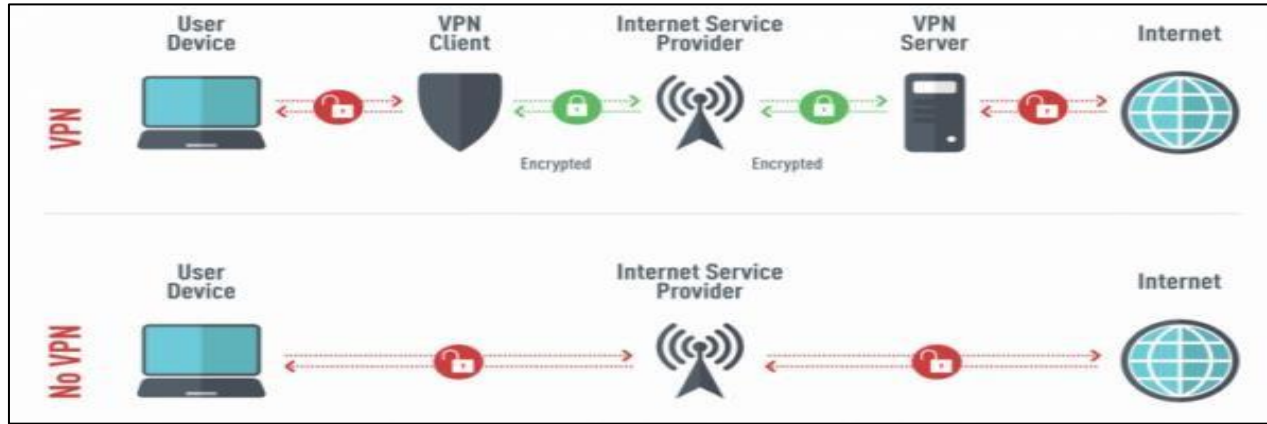


- Table summarizes the research projects on VM migration planning.

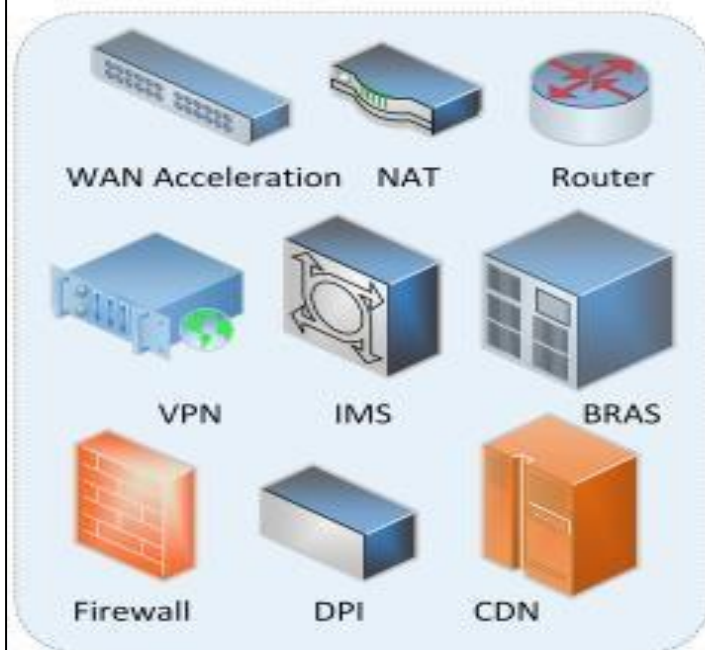
Project	Objectives	Approach/Technique	Evaluation
Bari et al. [37]	Finding sequence of migrations while migration time is minimized	VM migration – Heuristic	Simulation
Ghorbani et al. [38]	Finding sequence of migrations while imposing bandwidth guarantees	VM migration – Heuristic	Simulation
Li et al. [39]	Finding sequence of migrations and destination hosts to balance the load	VM migration – Heuristic	Simulation
iAware [40]	Minimization of migration and co-location interference among VMs	VM migration – Heuristic	Prototype/ Simulation

c. Virtual Network Functions Management

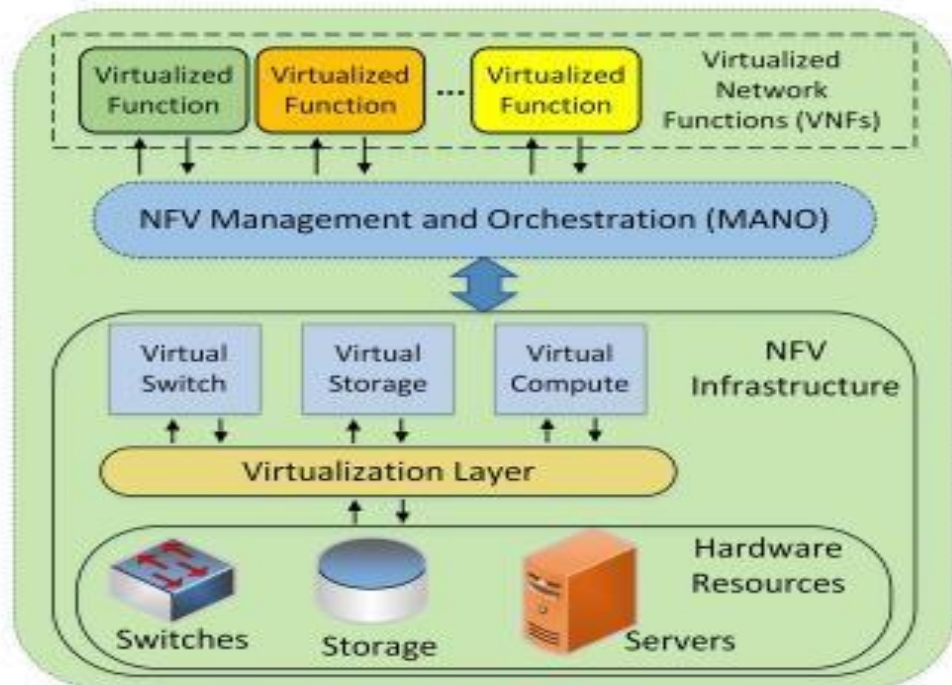
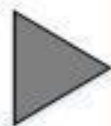
- NFV is an emerging paradigm where network functions such as **Firewalls**, **Network Address Translation** (NAT) and **Virtual Private Networks** (VPNs) are virtualized and divided up into multiple building blocks called **Virtualized Network Functions (VNFs)**.



Traditional Network Appliances

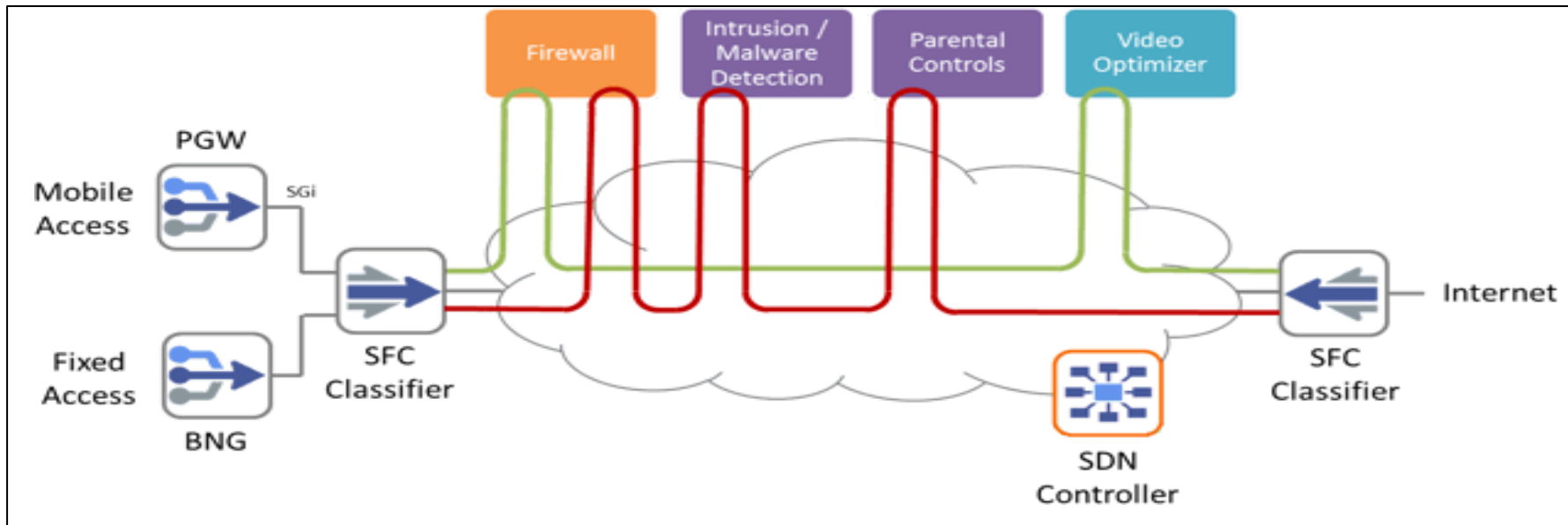


CDN: Content Delivery Network
IMS: IP Multimedia Subsystem
NAT: Network Address Translation
VPN: Virtual Private Network
BRAS: Broadband Remote Access Server
DPI: Deep Packet Inspection

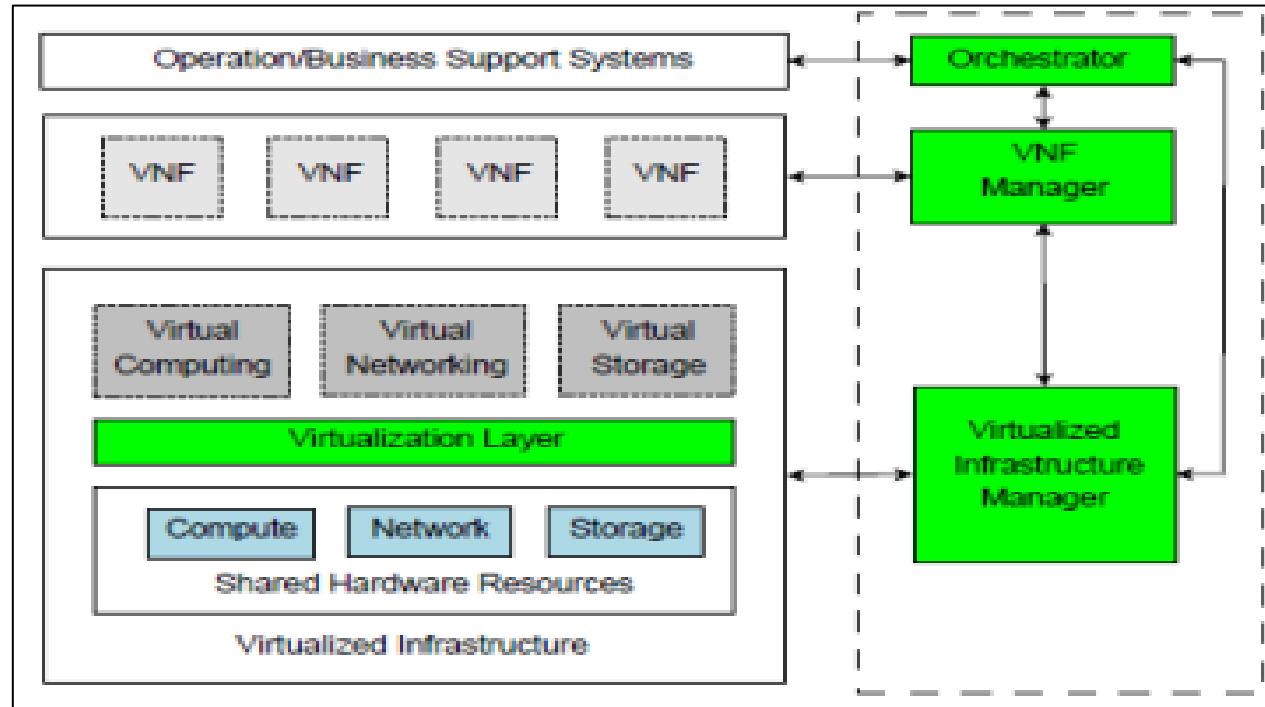


Network Functions Virtualization (NFV)

- VNFs are often chained together and build **Service Function Chains** (SFC) to deliver a required network functionality.



- **Han et al. (Network function virtualization: Challenges and opportunities for innovations)** present a comprehensive survey of key challenges and technical requirements of NFV where they present an architectural framework for NFV.
- They focus on the **efficient Implementation**, **placement** and **migration of VNFs**, and **network performance**.

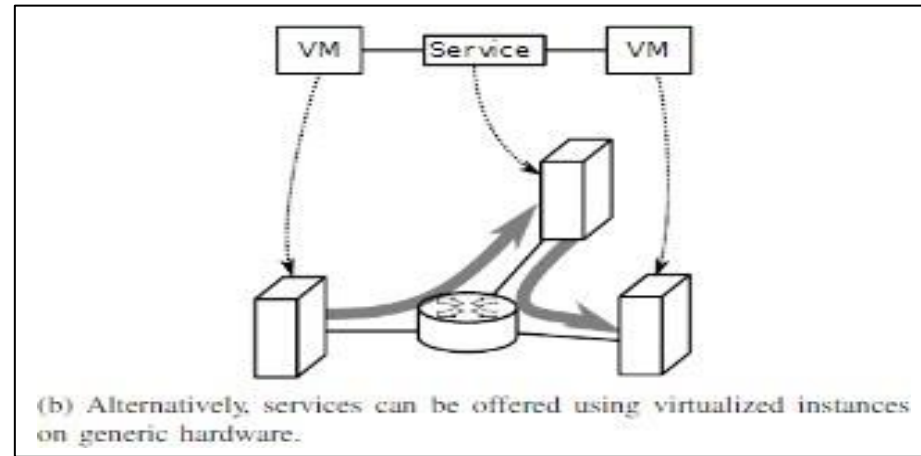
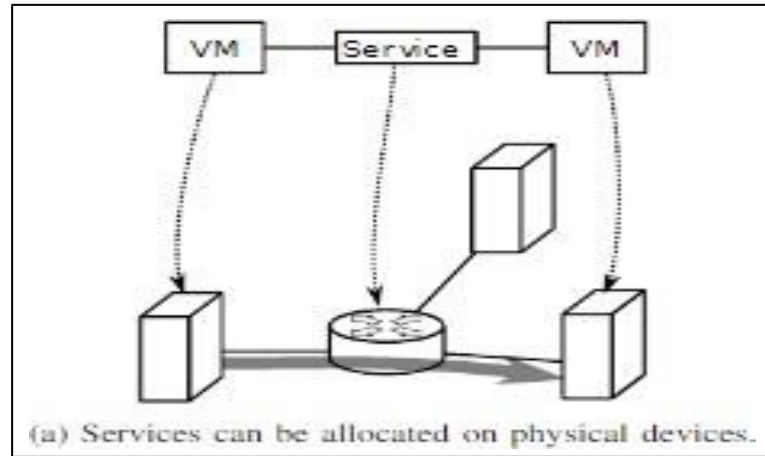


Imagine you are managing a food delivery system where:
Virtual Network Functions (VNFs) are like food preparation stations (e.g., pizza, burgers, desserts) that handle specific tasks.
The goal is to place these stations in the right locations (restaurants) and make sure they work efficiently to deliver food quickly to customers.

The study focuses on:

- **Efficient Implementation:** Setting up these stations (VNFs) so they work smoothly and deliver orders on time.
- **Optimal Placement:** Deciding where to place these stations (near busy areas or main roads) to reduce delivery time.
- **Smart Migration:** If demand changes (e.g., more people order pizza in a different area), they plan to move the pizza station to a better location.
- **Improving Network Performance:** Ensuring delivery routes (networks) are fast and reliable, so customers get their food without delays.

- **Moens and Turck (VNF-P: A model for efficient placement of virtualized network functions)** proposed a **VNF-P model** for efficient placement of VNFs.
- They propose a NFV burst scenario in a hybrid scenario in which the **base demand for network function** service is handled by **physical resources** while the **extra load** is handled by **virtual service instances**.

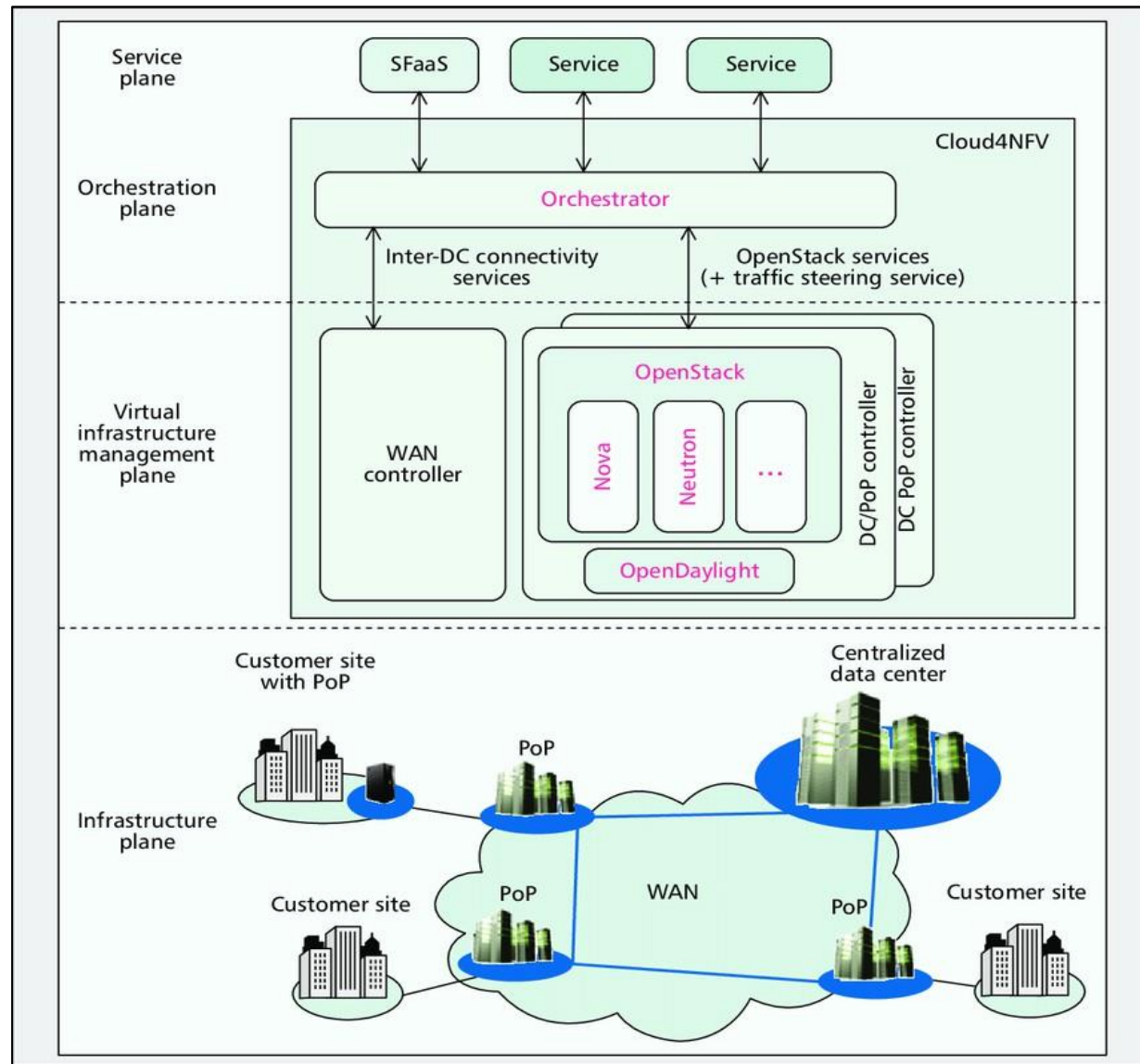


Imagine a coffee shop where:

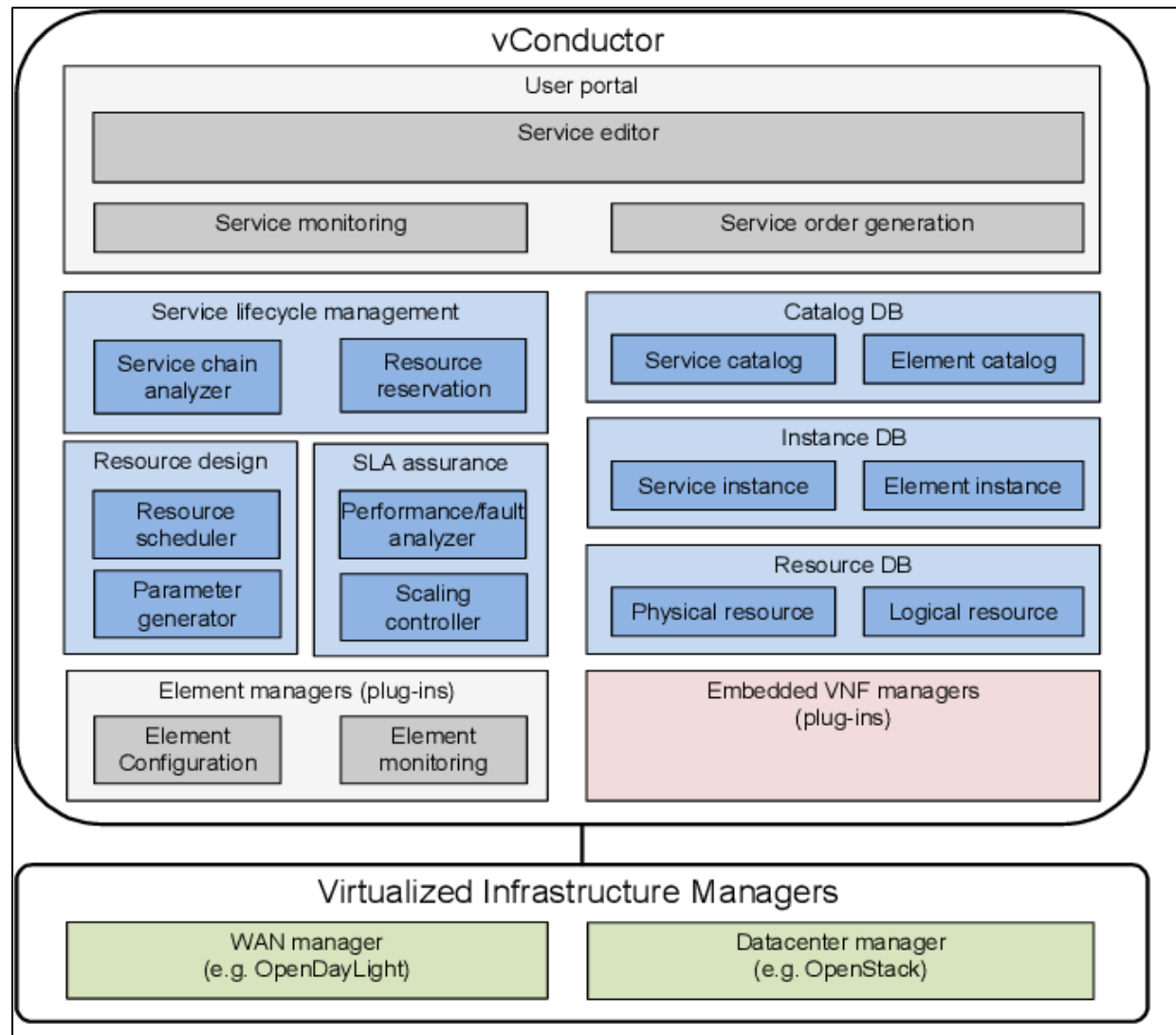
Base Demand: The regular number of customers is handled by the permanent staff (physical resources).

Extra Load (Burst): During busy times (like a morning rush), additional temporary workers (virtual service instances) are brought in to help serve customers quickly.

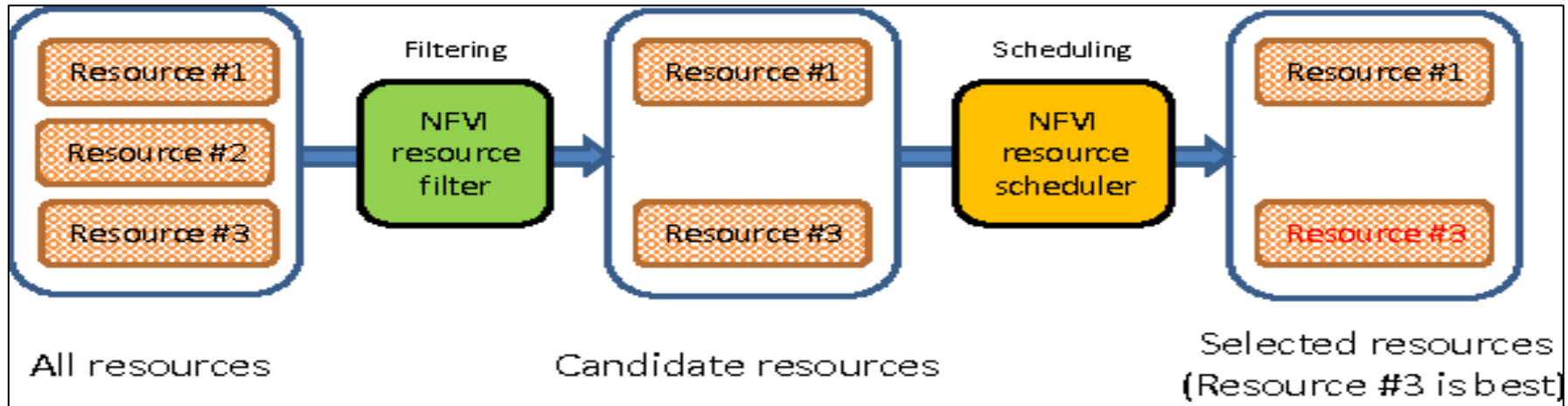
- **J. Soares et al.** (Cloud4NFV: A platform for virtual network as a functions platform) following standards European the NFV by the Telecommunications Standards Institute (ETSI) to build network function as a service using a cloud platform.
- Its VNF Orchestrator exposes RESTful APIs, allows VNF deployment.



- **W. Shen et al.** (**vConductor: An NFV management solution for realizing virtual services**)
- NFV management system for the end-to-end network virtual services.
- vConductor has simple graphical user interfaces (GUIs) for automatic provisioning of virtual network services and supports the management of VNFs and existing physical network functions.



- **Yoshida et al. (MORSA: A multi-objective resource scheduling algorithm for NFV infrastructure)** proposed as part of vConductor using virtual machines (VMs) for **building NFV infrastructure** in the presence of conflicting objectives that involve stakeholders such as users, cloud providers, and telecommunication network operators.



Imagine running a theme park where multiple groups have different goals:

Visitors (Users) want short wait times and smooth rides.

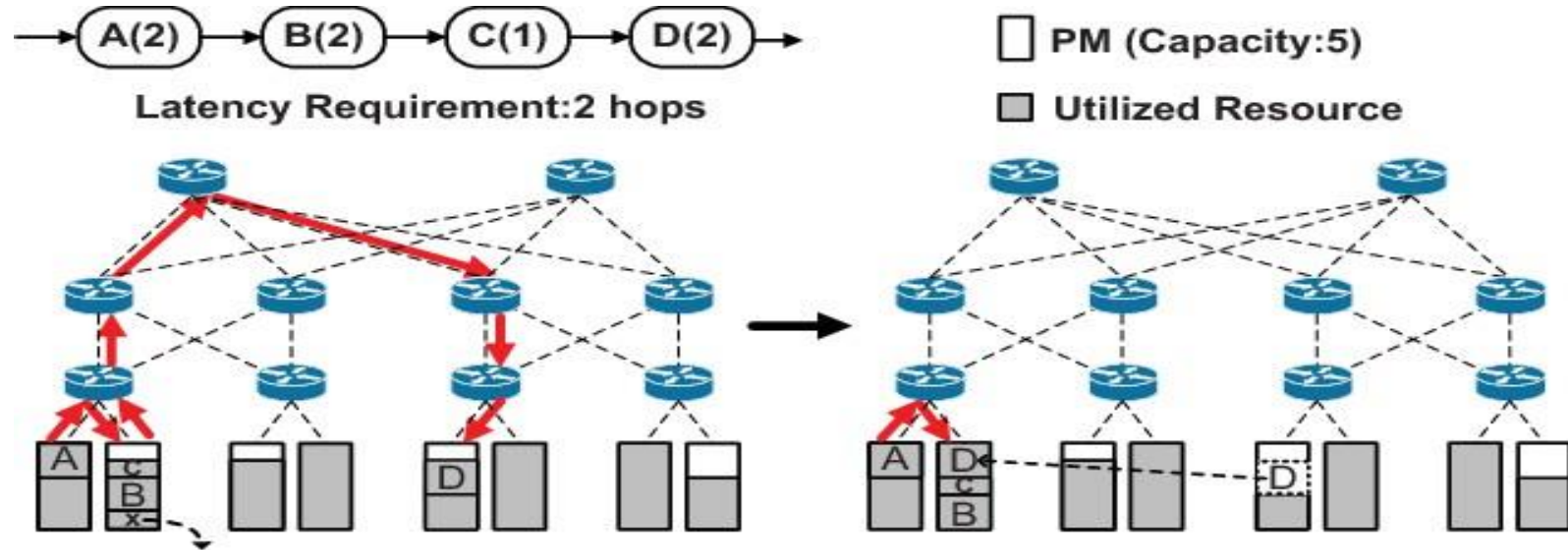
Park Managers (Cloud Providers) want to maximize profits by using resources efficiently.

Ride Operators (Telecom Network Operators) want to ensure rides are safe and well-maintained.

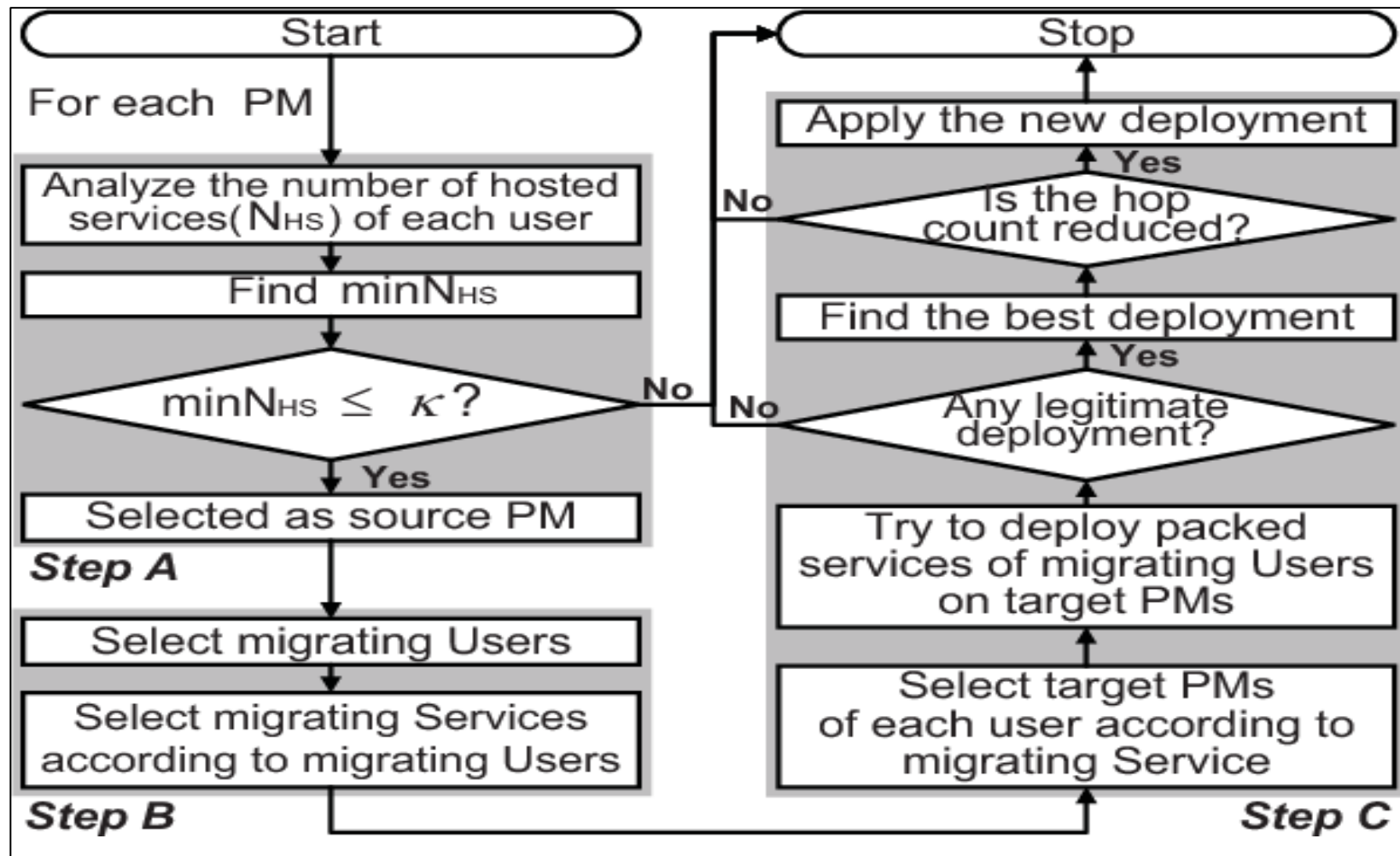
Now, Yoshida et al.'s **MORSA** is like a scheduling system that balances these conflicting goals:

- It assigns visitors (users) to the right rides (VMs) based on their preferences.
- It ensures the park resources (NFV infrastructure) are used efficiently to handle the crowd.
- It considers safety and maintenance schedules (telecom operators' needs) while keeping things running smoothly.

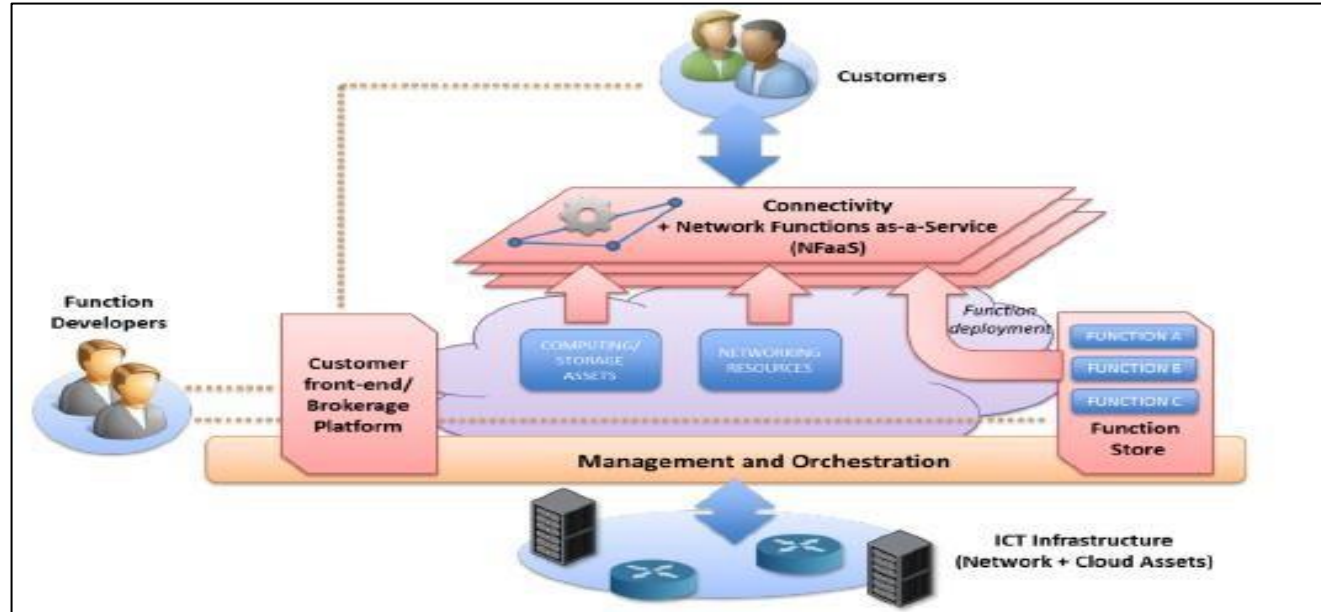
- **Y. F. Wu et al.** (TVM: Tabular VM migration for reducing hop violations of service chains in cloud datacenters) proposed to **reduce the number of hops** (network elements) in service chains of network functions in cloud data centers.
- They use **VM migration** to reduce the number of hops the flow should traverse to satisfy SLAs.

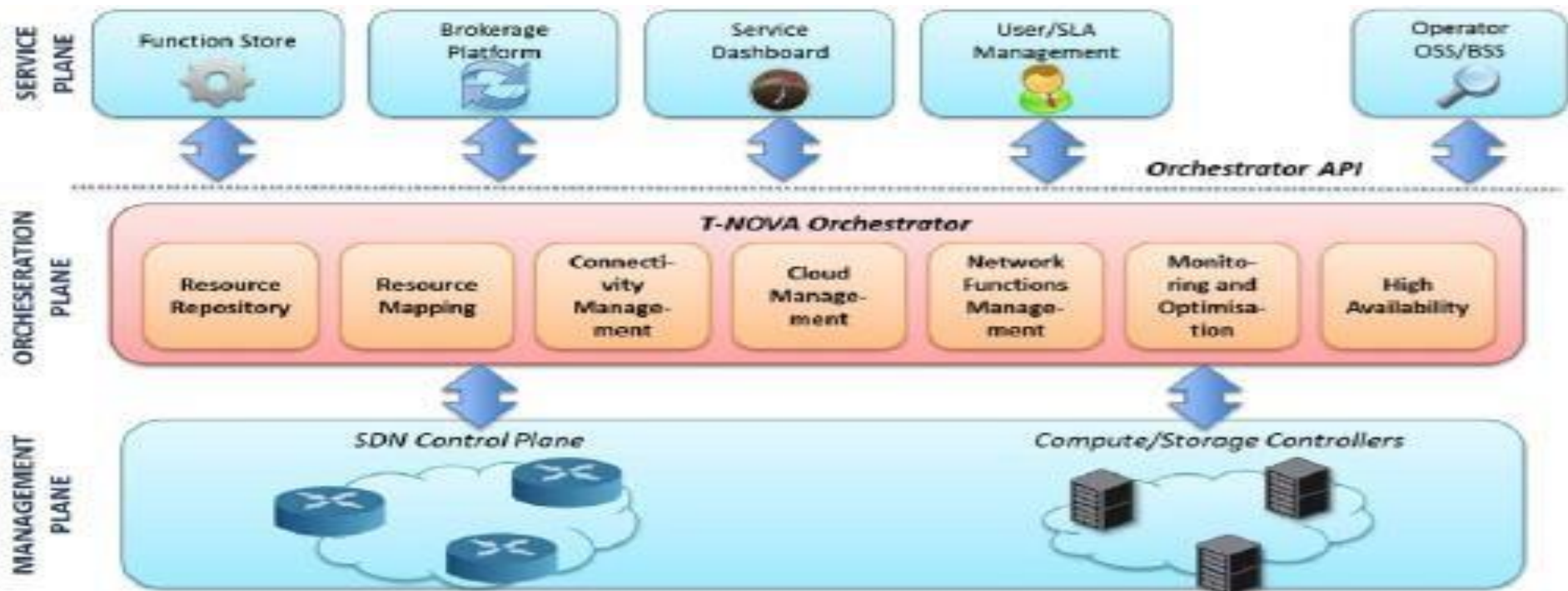


- Flowchart of TVM

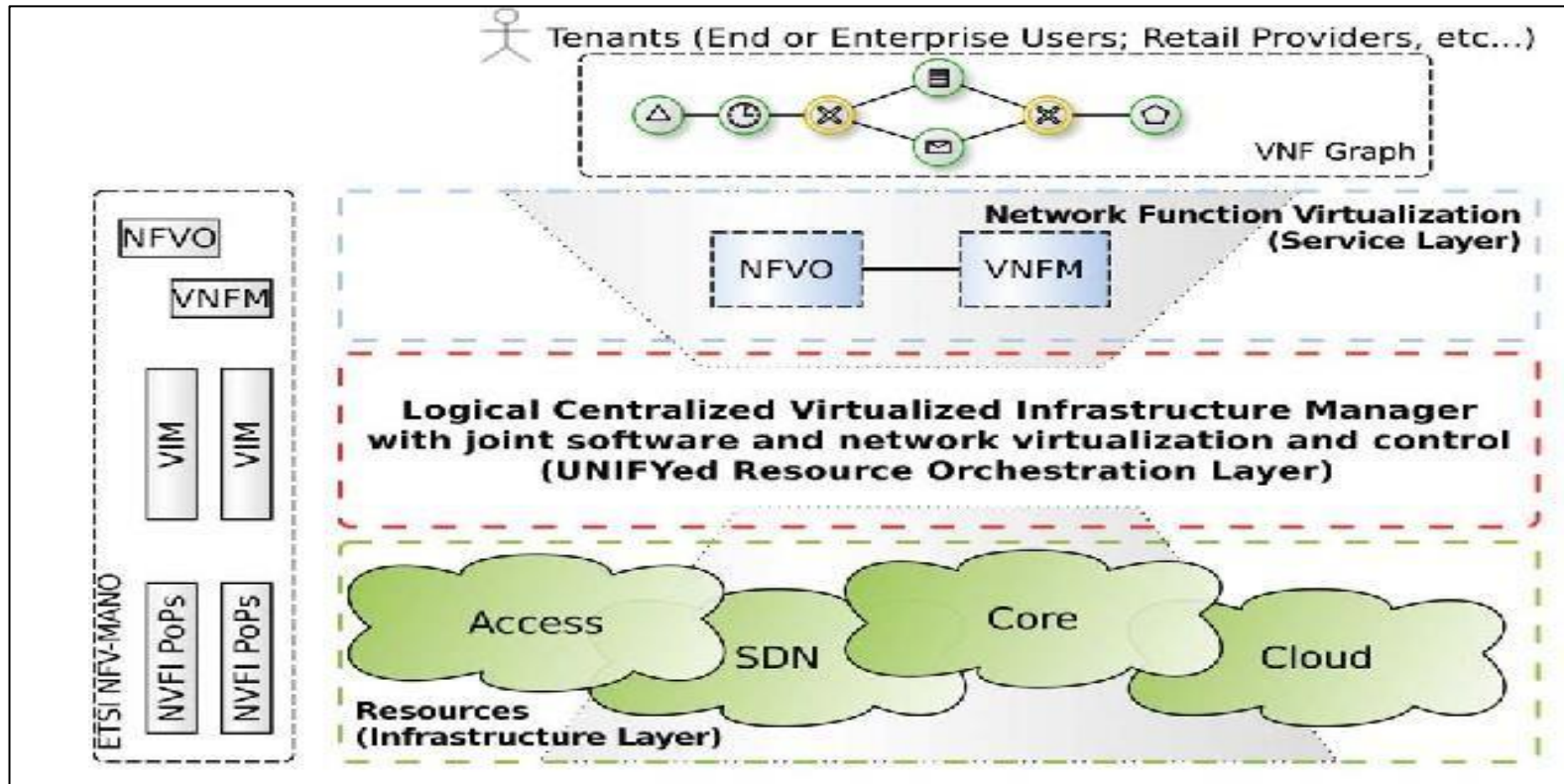


- **Pai et al. (SLA-driven ordered variable-width windowing for service-chain deployment in SDN datacenters)** is a heuristic proposed to address the same problem, however, using initial static placement.
- **G. Xilouris (T-NOVA: A marketplace for virtualized network functions)** the EU-funded T-NOVA project aims to realize the **NFaaS** (Virtual Network Function as a Service) **concept**.
- It has designed and implemented integrated management and orchestrator platforms for the automated provisioning management, monitoring, and optimization of VNFs.



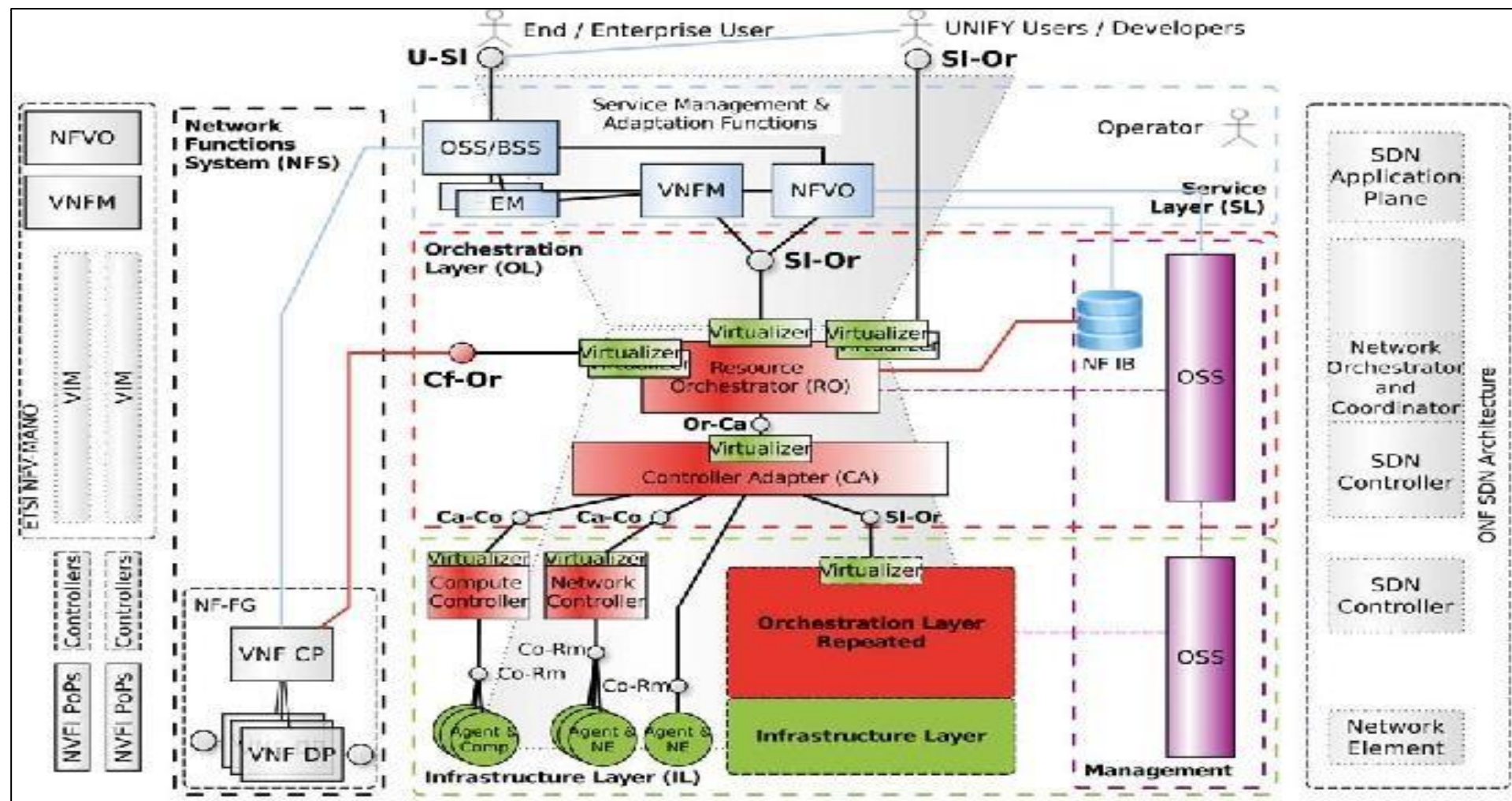


- **B. Sonkoly et al. (UNIFYing cloud and carrier network resources: an architectural view)** proposed another **EU-funded FP7 project** that is aimed at supporting **automated, dynamic service creation** based on a fine-granular SFC model, SDN, and cloud virtualization techniques.



UNIFY Objectives: Unifying cloud and network resources

UNIFY Overarching Architecture

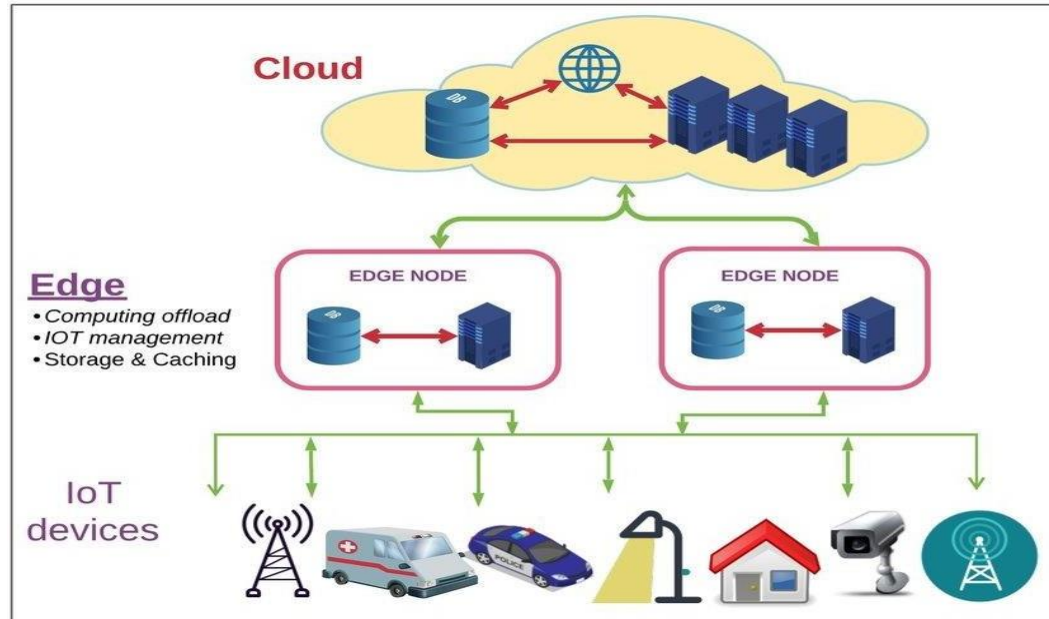


- Table summarizes Virtual network functions management projects.

Project	Objectives	Approach/Technique
VNF-P	Handling burst in network services demand while minimizing the number of servers	Resource allocation - Integer linear programming (ILP)
Cloud4NFV	Providing network function as a service	Service provisioning – Framework design
vConductor	Virtual network services provisioning and management	Service provisioning – Framework design
MORSA	Multi objective placement of virtual services	Placement - Multi-objective genetic algorithm
TVM	Reducing number of hops in service chain	VNF migration - heuristic
SOVWin	Increasing user requests acceptance rate and minimization of SLA violation	VNF placement - heuristic
Clayman et al.	Providing automatic placement of the virtual nodes	VNF placement - heuristic
T-NOVA	Building a marketplace for VNF	Marketplace – framework design
UNIFY	Automated, dynamic service creation and service function chaining	Service provisioning– framework design

Network Slicing Management in Edge and Fog

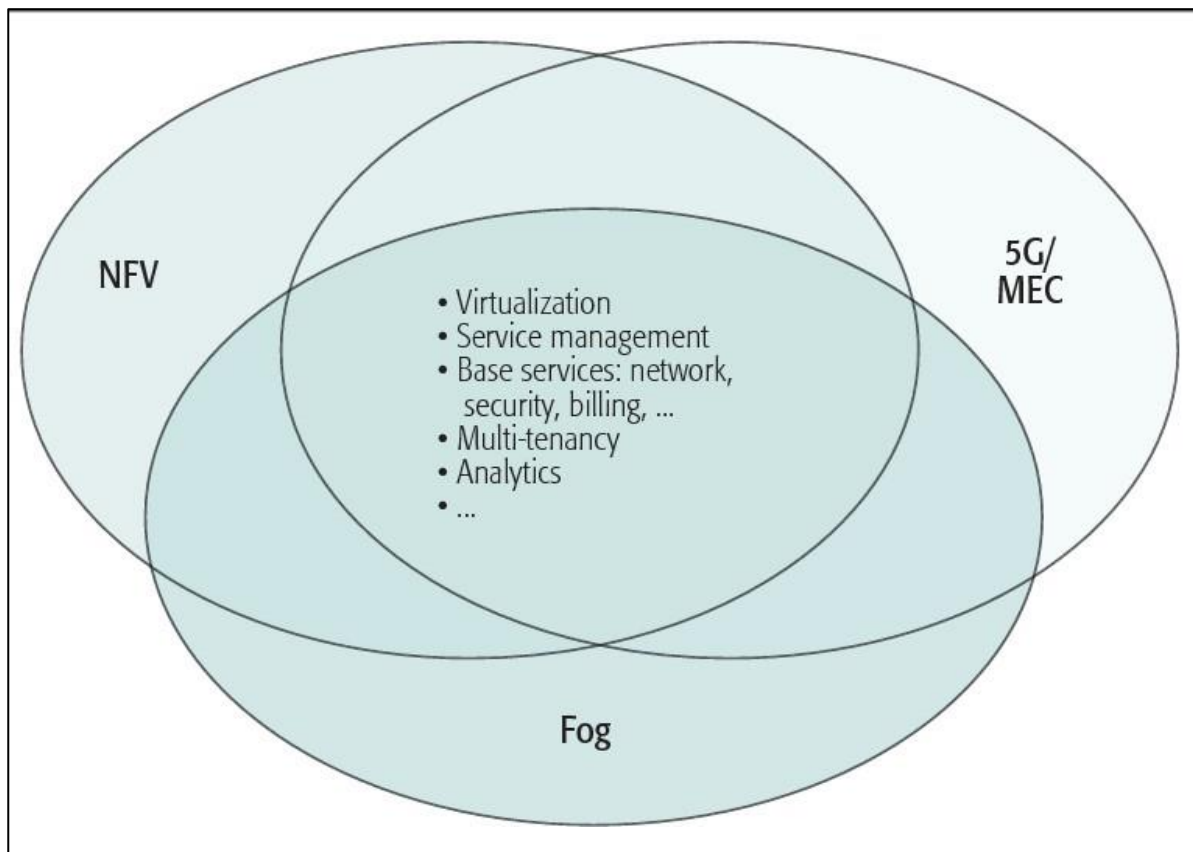
- Fog computing is a new trend in cloud computing that attempts to address the quality of service requirements of applications requiring real-time and low-latency processing.
- Fog acts as a middle layer between edge and core clouds to serve applications close to the data source, core cloud data centers provide massive data storage, heavy-duty computation, or wide area connectivity for the application.



- The key visions of fog computing is to add compute capabilities or general-purpose computing to edge network devices such as mobile base stations, gateways, and routers.
- SDN and NFV play key roles in prospective solutions to facilitate efficient management and orchestration of network services.
- Interaction between SDN/NFV and fog/edge computing is crucial for emerging applications in IoT, 5G, and stream analytics

- **Lingen et al.** (The unavoidable convergence of NFV, 5G, and fog: A model-driven approach to bridge cloud and edge) define a **model-driven and service-centric architecture** that addresses technical challenges of integrating NFV, fog, and 5G/MEC.

- They introduce an open **architecture based on NFV MANO** proposed by the European Telecommunications Standards Institute (ETSI) and aligned with the Consortium OpenFog (OFC) reference architecture that offers uniform management services of IoT spanning through cloud to the edge.



A smart traffic management system uses IoT sensors installed at traffic lights, vehicles, and road infrastructure. These sensors generate real-time data, such as vehicle counts, speeds, and traffic density. The goal is to optimize traffic flow and reduce congestion.

Cloud Layer:

- The cloud serves as a centralized data repository for historical traffic data and supports machine learning models to predict long-term traffic patterns.
- High-performance computing resources in the cloud analyze large-scale data for trends and insights.

Fog Layer:

- Fog nodes (e.g., roadside servers or gateways) process real-time traffic data locally.
- For instance, fog nodes can calculate optimal traffic light timings based on nearby congestion levels, reducing latency compared to cloud processing.

Edge Layer:

- Edge devices (e.g., cameras, IoT sensors) capture real-time data and send it to the fog layer.
- Minimal processing, such as filtering noisy data, is done here to ensure efficient communication with the fog.

Integration of NFV, 5G, and MEC:

NFV:

- Virtual network functions (e.g., real-time video analytics) are deployed dynamically on fog nodes to adapt to changing traffic conditions.

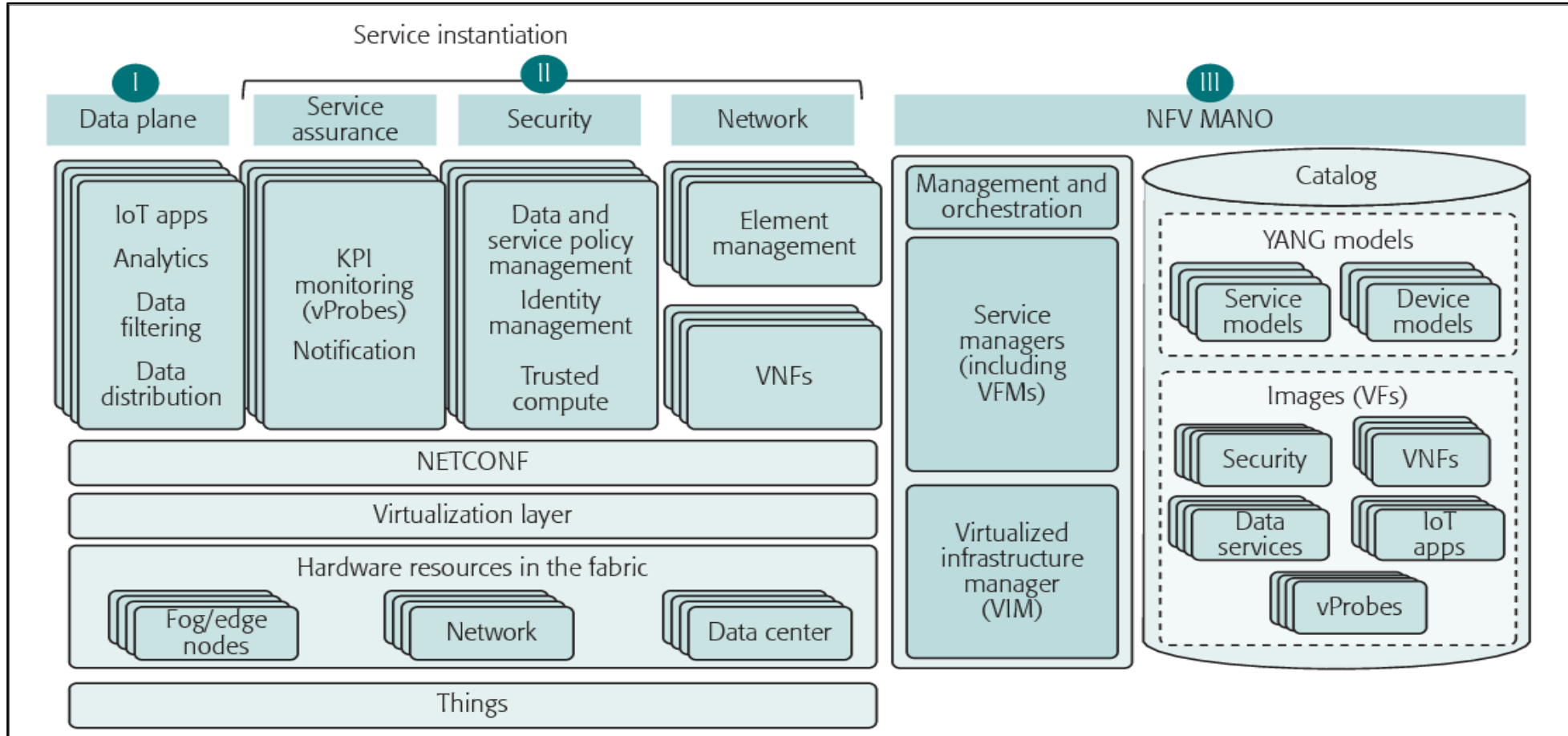
5G/MEC:

- Ultra-low latency of 5G ensures fast communication between edge devices, fog nodes, and vehicles.

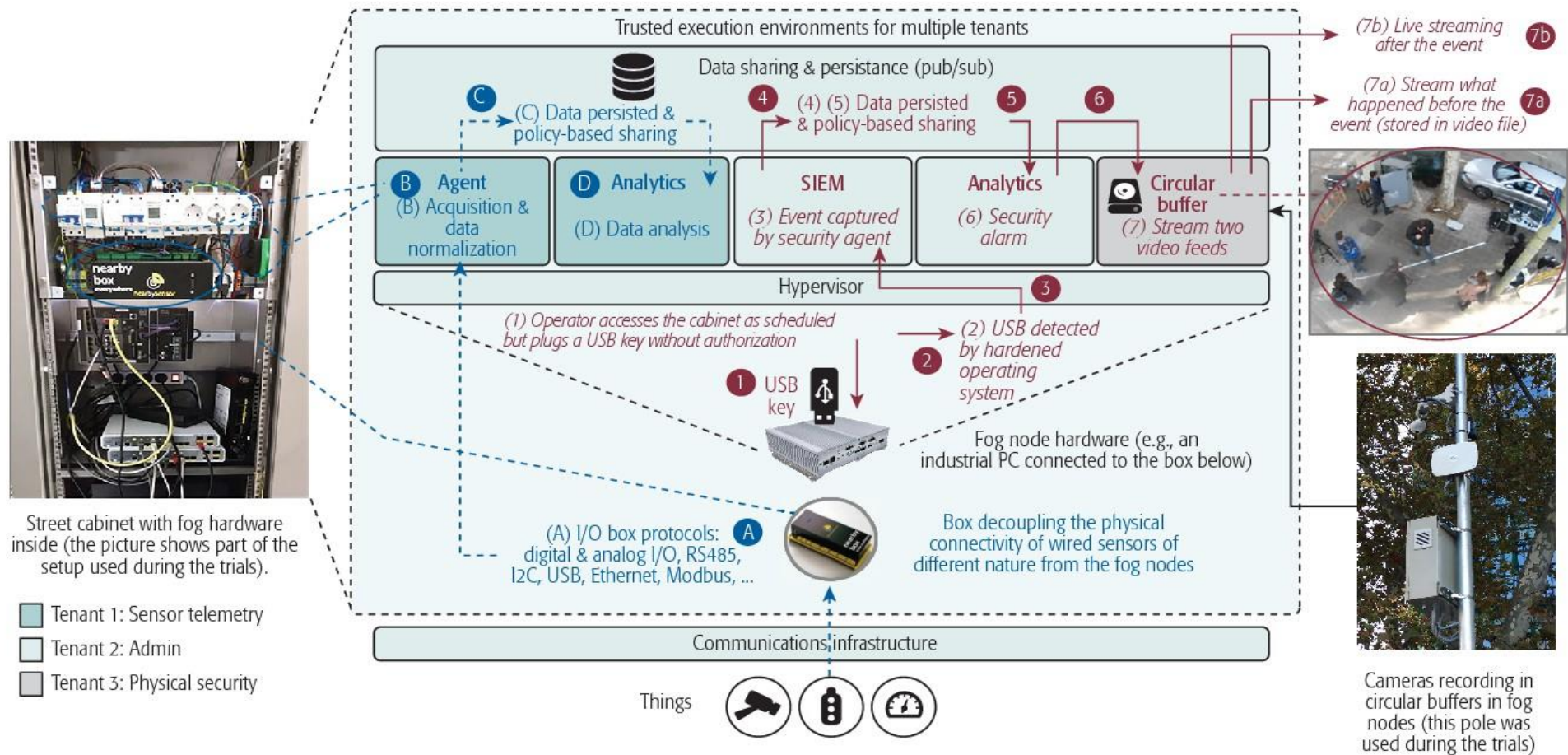
Uniform Management:

- ETSI NFV MANO and OFC frameworks enable seamless orchestration of these resources, ensuring consistent performance and reliability across all layers.
- Real-time traffic data is processed locally at the fog layer for immediate actions like adjusting traffic lights.
- Long-term predictions are updated and refined in the cloud.

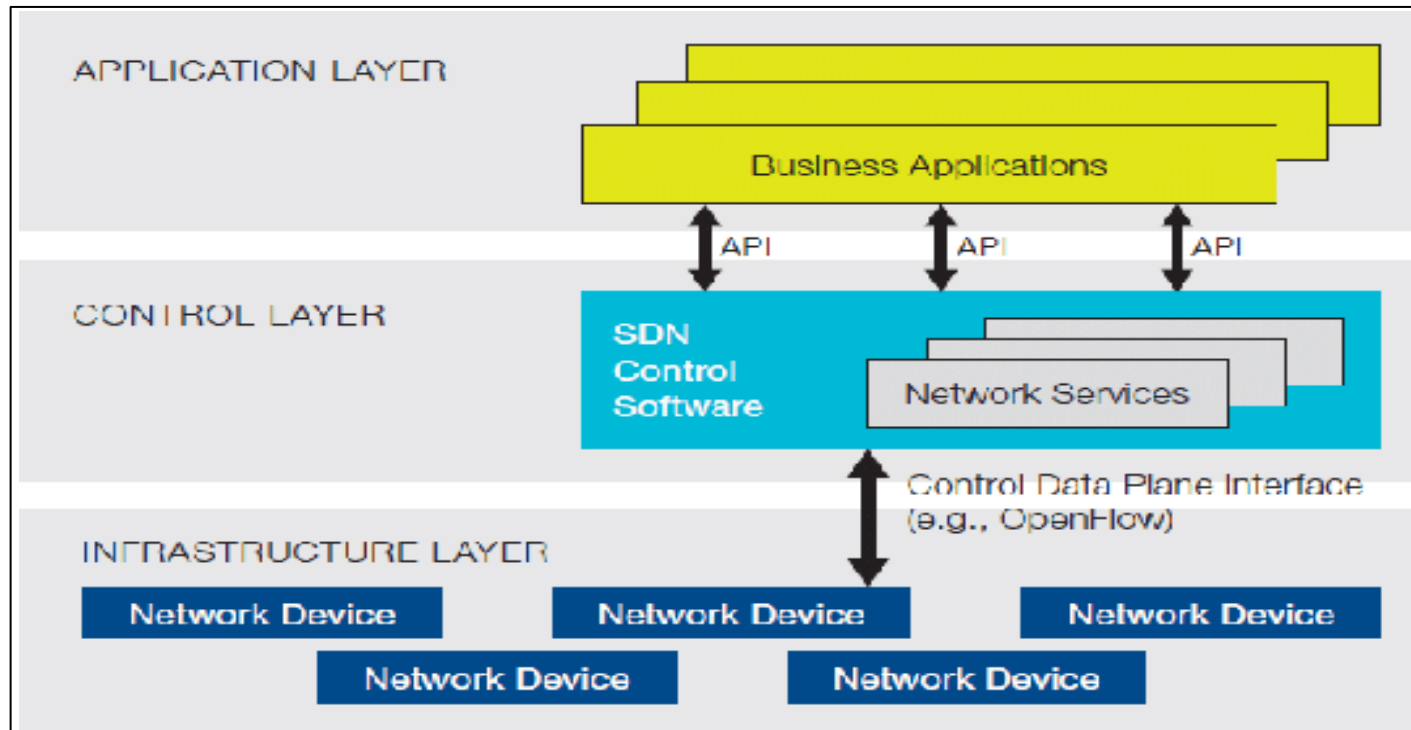
- A **two-layer abstraction model** along with IoT-specific modules and enhanced NFV MANO architecture is proposed to integrate cloud, network, and fog.



- They presented two use cases for physical security of fog nodes and sensor telemetry through **street cabinets** in the city of Barcelona.



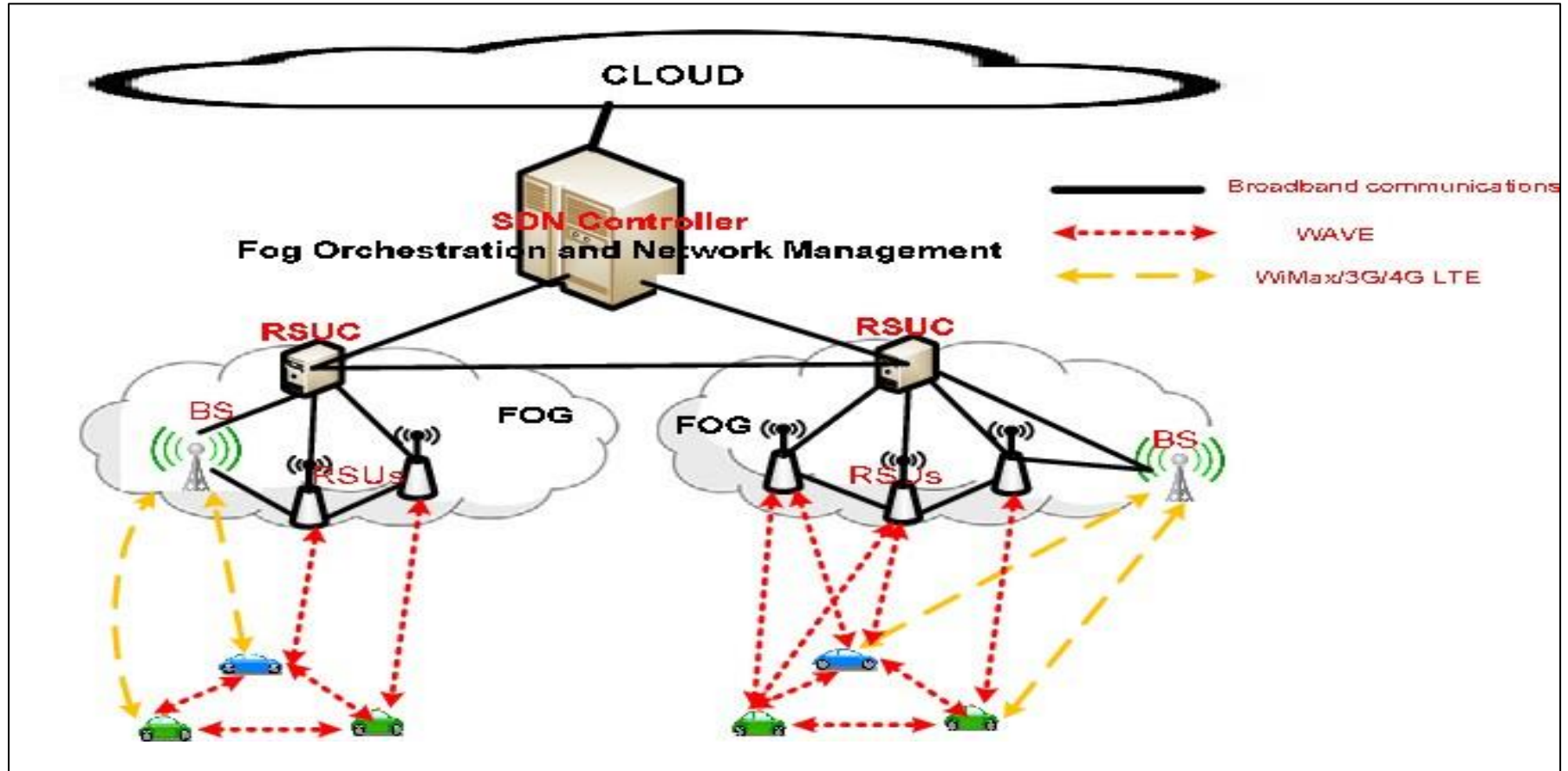
- **Truong et al.** (Software defined networking-based vehicular adhoc network with fog computing) proposed an **SDN-based architecture** to support fog computing.
- They have identified required components and specified their roles in the system.



Imagine a fleet of autonomous vehicles driving through a city.

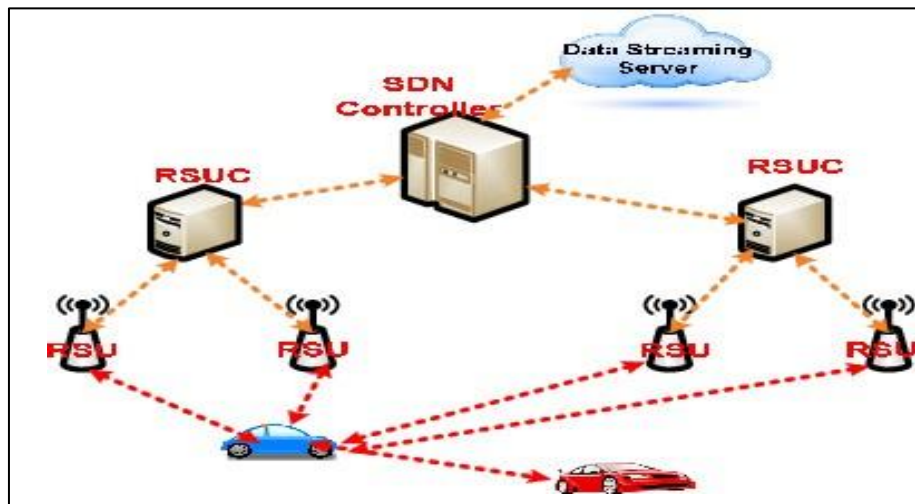
- Each vehicle generates data, such as speed, location, and environmental conditions (e.g., traffic lights, road conditions).
- Rather than sending all of this data to the cloud for processing, which could introduce delays, some data is processed locally at fog nodes (located at intersections).
- The SDN controller manages the traffic and determines which data should be processed locally at the fog nodes and which data needs to be sent to the cloud for further analysis.
- For example, if a vehicle detects an obstacle and needs to react quickly, the decision-making process (such as activating brakes) is done at the nearby fog node.
- On the other hand, historical data about the city's traffic patterns may be sent to the cloud for long-term analysis.

- They also showed how their system can provide services in the context of **Vehicular Adhoc Networks** (VANETs).



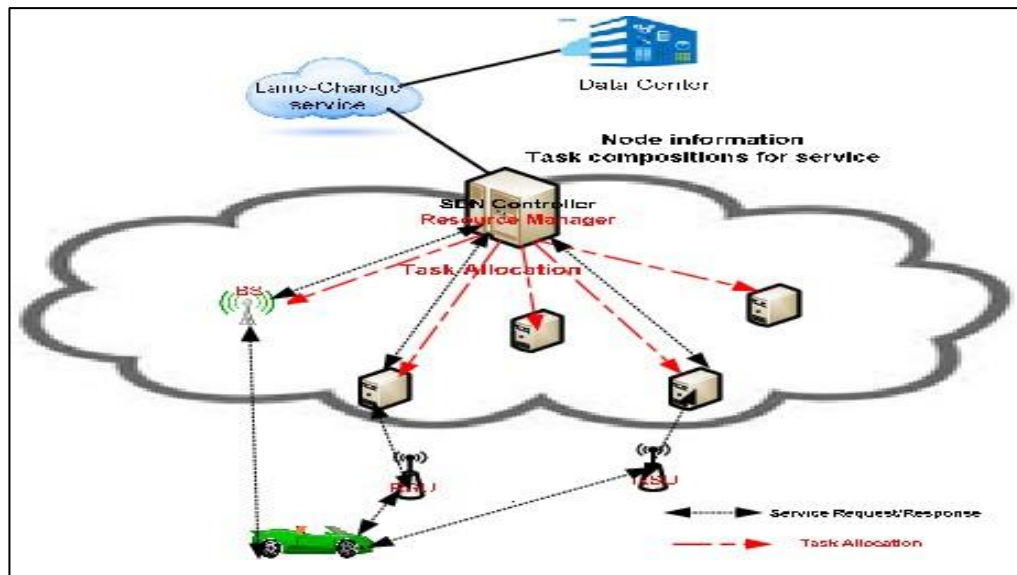
SDN-based VANET architecture leveraging Fog Computing

- They showed benefits of their proposed architecture using two use-cases in **data streaming** and **lane-change assistance services**.



Data Streaming Reference Scenario

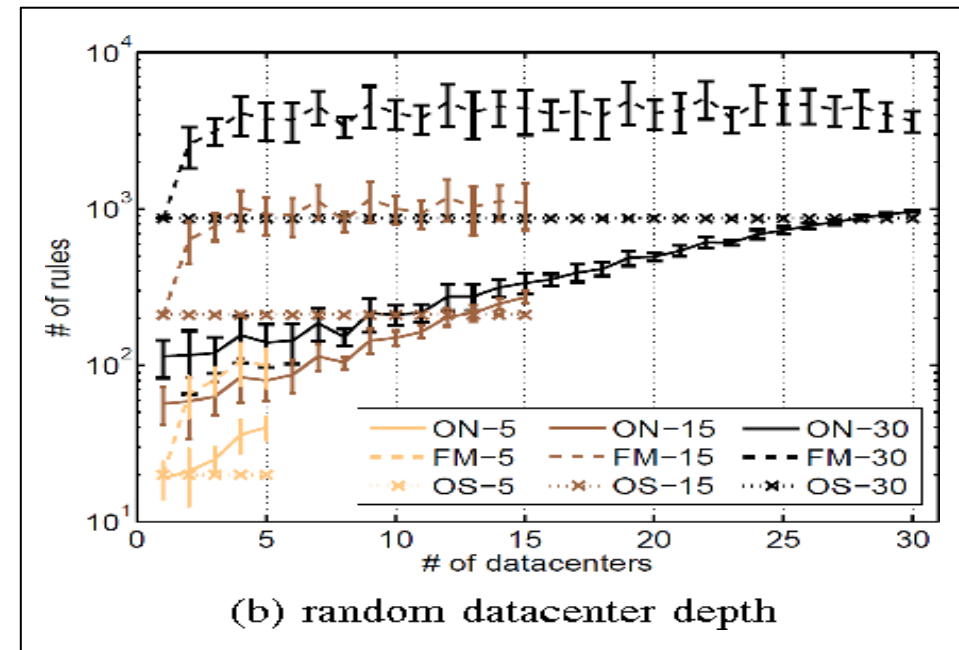
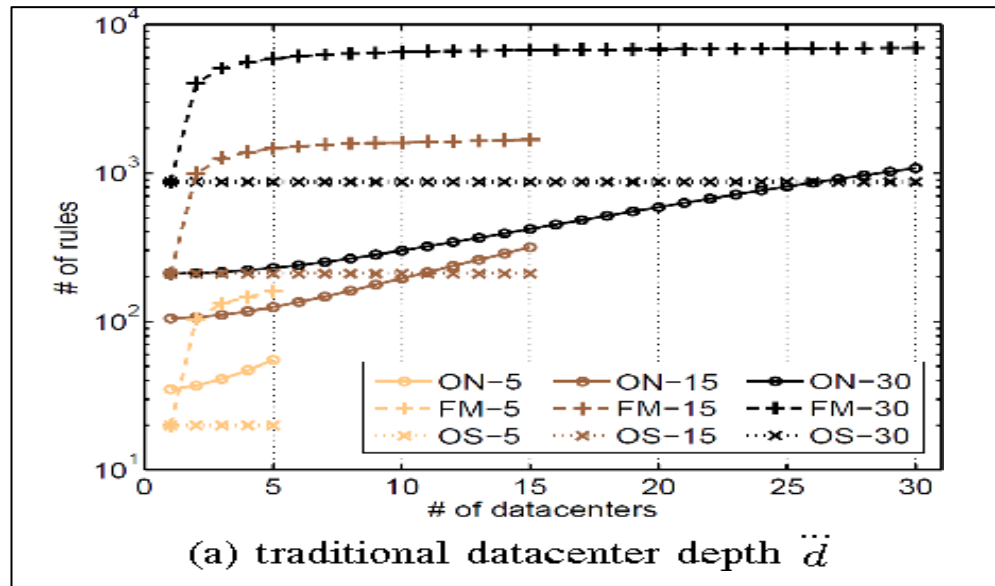
A car's camera streams live video of the road ahead, and the fog node at an intersection analyzes it for relevant traffic information (such as detecting vehicles or pedestrians). The fog node processes this data locally and only sends important events (like accidents or unusual traffic patterns) to the cloud for further analysis.



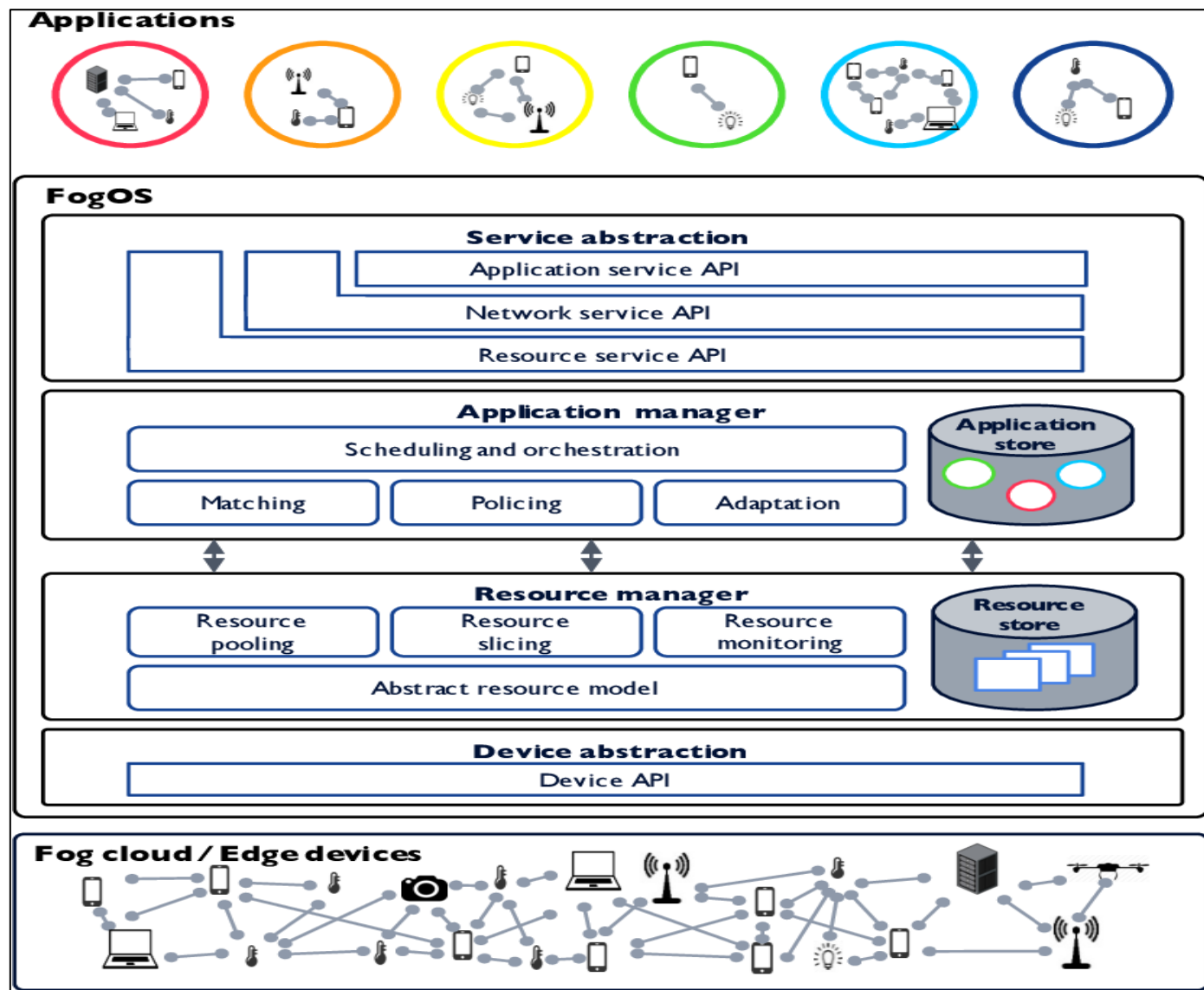
Lane Change Service in FSND

A vehicle decides to change lanes, but before doing so, it needs to check if there are any vehicles in the next lane or if the lane is blocked. The fog node nearby processes data from surrounding vehicles and road sensors in real time and informs the vehicle's system whether it's safe to change lanes. If there is a risk (e.g., a car is too close), the system warns the driver or takes corrective action, like adjusting the vehicle's speed.

- **Bruschi et al. (A scalable SDN slicing scheme for multi-domain fog/cloud services)** proposed a **network slicing scheme** for supporting multi domain fog/cloud services.
- They propose SDN-based network slicing scheme to build an overlay network for geographically distributed Internet services using non-overlapping OpenFlow rules.
- Their experimental results show that the number of unicast forwarding rules installed in the overlay network significantly drops compared to the fully meshed and OpenStack cases.

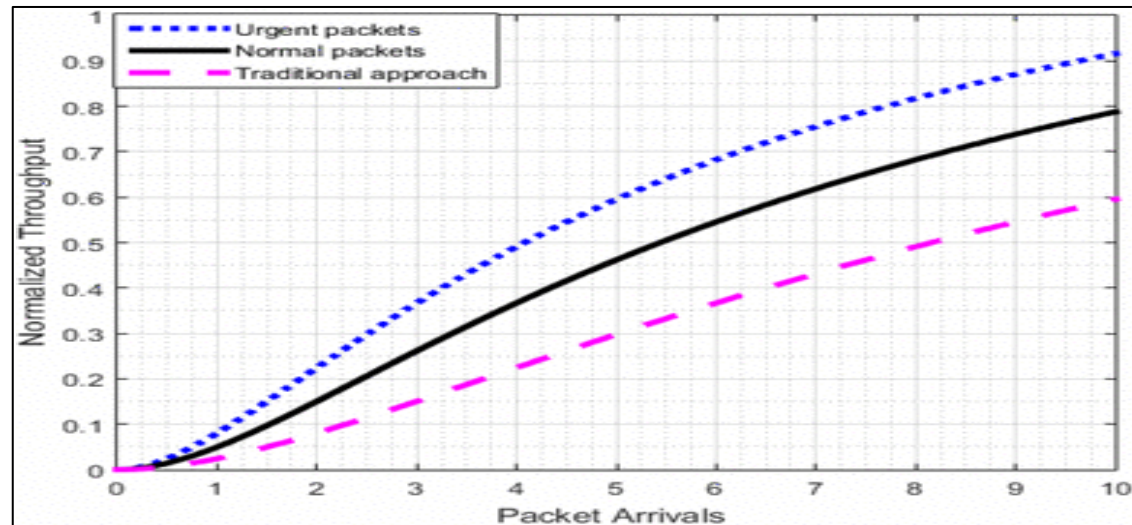
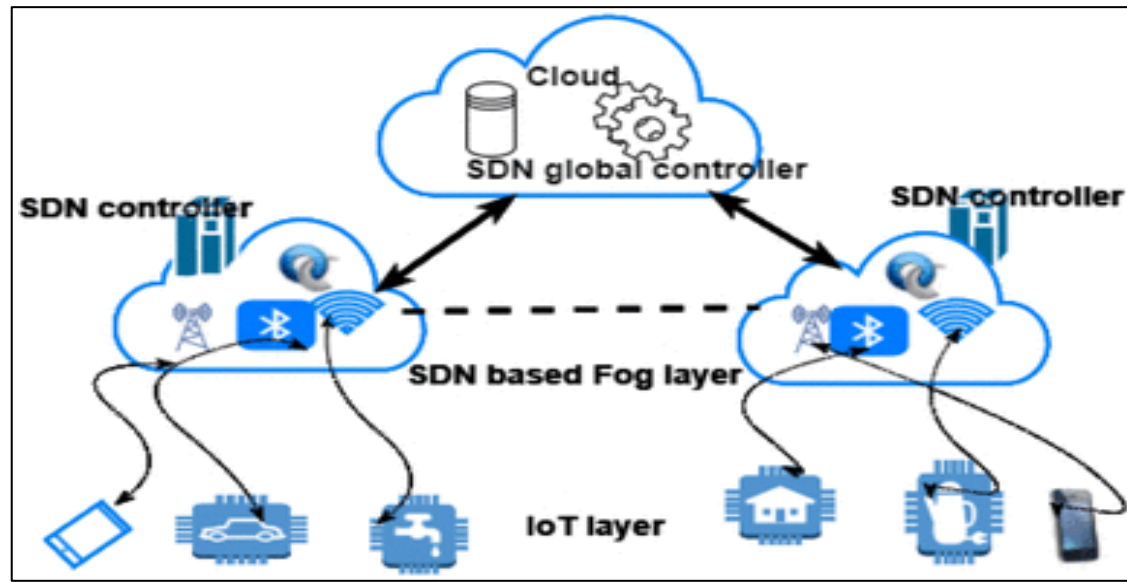


- **Choi et al.** (A fog operating system for user-oriented IoT services: Challenges and research directions) proposed a fog operating system architecture called FogOS for IoT services.

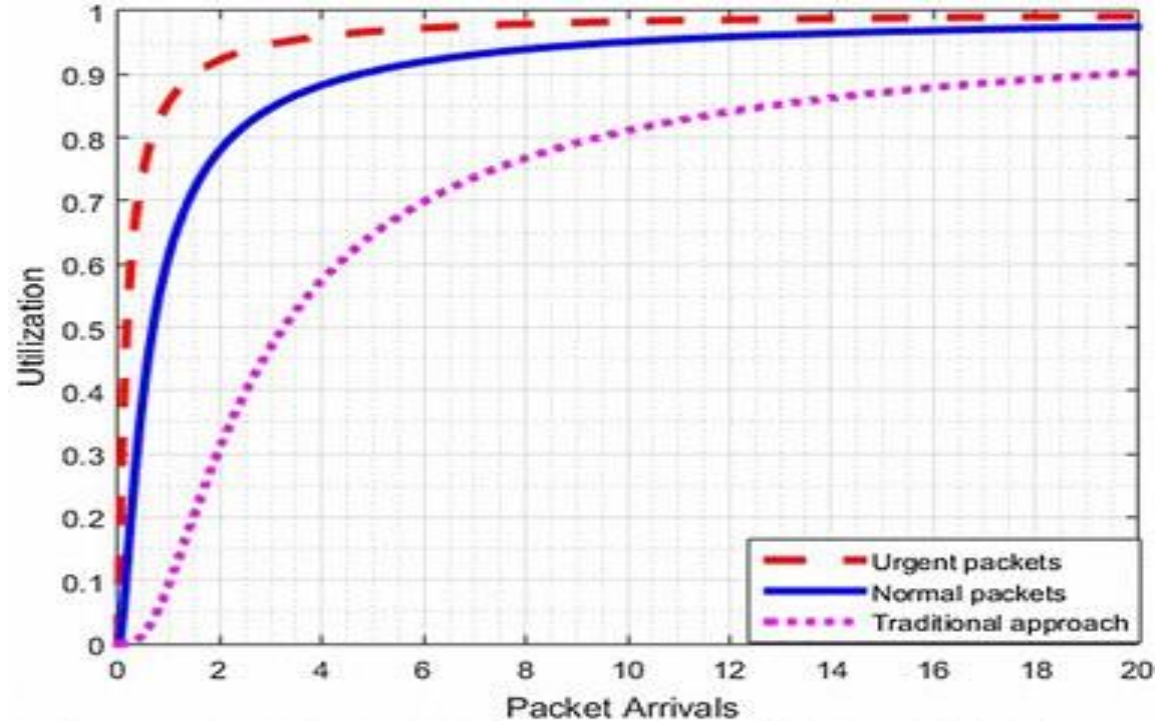


- They identified four main challenges of fog computing:
 - **Scalability** for handling significant number of IoT devices
 - **Complex inter-networking** caused by diverse forms of connectivity, e.g., various radio access technologies
 - **Dynamics and adaptation** in topology and quality of service (QoS) requirements
 - **Diversity and heterogeneity** in communications, sensors, storage, and computing powers, etc
- Based on these challenges, their proposed architecture consists of four main components:
 - 1. Service and device abstraction**
 - 2. Resource management**
 - 3. Application management**
 - 4. Edge resource: registration, ID/addressing, and control interface**

- **Diro et al. (Differential flow space allocation scheme in SDN based fog computing for IoT applications)** proposed a mixed SDN and fog architecture that gives priority to critical network flows while taking account into among other flows in the fog-to-things communication to satisfy QoS requirements of heterogeneous IoT applications.
- They intend to satisfy QoS and performance measures such as packet delay, lost packets, maximized throughput.



- Results show that their proposed method can serve critical and urgent flows more efficiently while allocating network slices to other flow classes.



Flow space utilization

Imagine a smart traffic management system in a city that uses IoT devices. Sensors are installed on roads to monitor traffic congestion, detect accidents, and manage traffic lights. Cameras provide real-time footage, and emergency vehicles (like ambulances) are equipped with communication devices to request priority clearance at intersections.

Problem:

The system needs to ensure:

- Critical traffic flow is prioritized – For example, ambulances must always be given a green signal when approaching an intersection.
- Other data flows (e.g., CCTV footage, traffic sensor data) are also managed effectively without causing delays.
- The overall system must meet QoS (Quality of Service) requirements, like low packet delay, minimized packet loss, and maximized throughput.

SDN Role:

- SDN controllers dynamically manage and route network traffic.
- They recognize emergency vehicle communication as critical flows and prioritize these over other data flows, like CCTV footage.
- For example, if an ambulance sends a signal to request green-light clearance, the SDN controller instantly adjusts traffic light settings and ensures its data is transmitted with minimal delay.

Fog Computing Role:

- Fog nodes are placed near the roads (closer to the IoT devices) to process data locally without relying solely on a distant cloud.
- This ensures quicker responses.
- For instance, if a road sensor detects a traffic jam, the fog node processes this data locally and communicates with nearby traffic lights to redirect traffic in real-time.